

No-clustering bootstrap screen $\text{Sharpe} > 0.5$ (OOS 2023+)

Directed factor-mimicking portfolio catalogue

Abstract

This catalogue constructs one FMP for each configured field or composite and compares all factors on one factor- and control-complete monthly window. Factors are indexed by reported raw Sharpe, highest first. Direction is frozen upstream; this report performs no return-based sign selection.

Methodology

Signals are observed at month-end, earn the next calendar month's simple return, and refresh monthly. Every factor must independently pass breadth, history, coverage, and non-degeneracy gates; none is silently dropped. One factor- and control-complete window fixes the reported Sharpe order and feeds all reported analyses. Displayed signal signs define the pre-direction composite; the reported Direction then orients the whole FMP. Risk Loadings include HAC t-statistics; Return Contributions are loading times annualized control mean, plus alpha and realized total. Rank diagnostics use 0/1/3/6/12/36-month horizons; available-history statistics are diagnostic only. Return Uncertainty uses paired circular stationary-bootstrap draws of the common monthly panel. It measures sampling and path uncertainty, not out-of-sample generalisation.

Construction Configuration

Setting	Effective value
Factors	21
Universe	shared security screen; refreshed monthly
Formation filters	price $\geq$ None; 21d dollar volume $\geq$ none
Market-cap screen	top 90.0% of names monthly
Signal preparation	within-month winsorize, normalize, then demean
Composite	mean available normalized components; factor-specific coverage floor
Construction	one blanket profile
Rebalance / return	month-end / next calendar month simple return
Costs	effective configured costs; default 0 bps
Minimum sample	20 stocks/month; 24 common months; 90% factor coverage
Comparison window	2023-01-01 to 2026-05-01; factor- and control-complete
Winsor settings	1%-99%
Controls	MKT, SMB, VAL, MOM, BAB, QMJ
Return uncertainty	paired stationary bootstrap; 2000 draws; expected block 12 months; seed 0
Catalogue note	Each of the 21 admitted signals is one standalone factor. No clustering, PCA, varimax, or post-screen factor pruning is applied. Admission requires at least 70% of fixed-orientation 2012–2022 stationary-bootstrap Sharpes to exceed 0.5.

Returns PCA

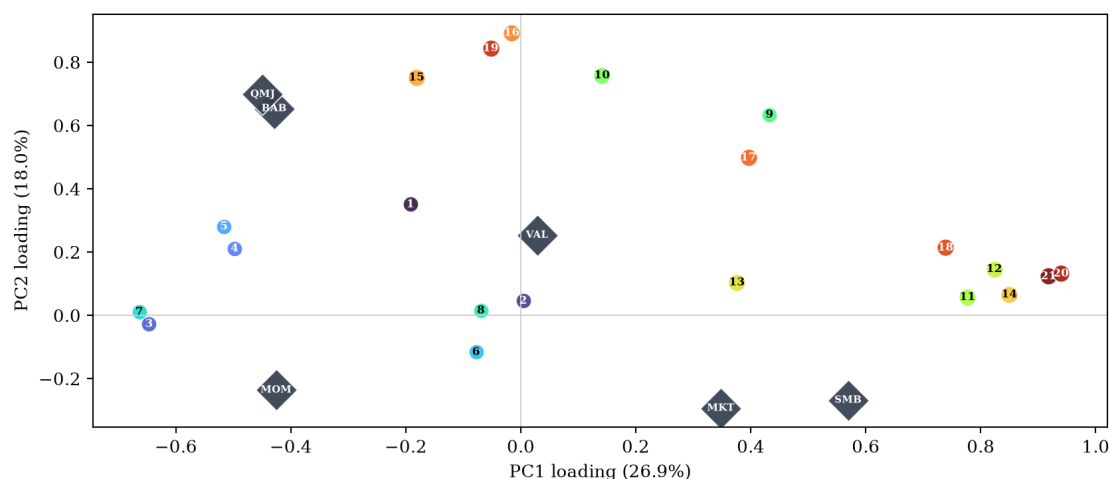


Figure: catalogue factors are numbered by reported-Sharpe index; attribution controls are named; $N=41$ common months.

Index

Idx	Factor	Dir.	N
1	share_turnover_21d	+1	41
2	accruals_noa_proxy	+1	41
3	momentum_21_252_vol_63d	+1	41
4	eps_growth_yoy	+1	41
5	net_income_growth_yoy	+1	41
6	ret_zscore_63d	+1	41
7	momentum_21_252	+1	41
8	resid_ret_252v_1260c_SPY	+1	41
9	dpo	+1	41
10	asset_turnover	+1	41
11	stoch_k_14d	+1	41
12	bb_pctb_20d	+1	41
13	accruals_total	+1	41
14	rsi_14d	+1	41
15	debt_to_assets	+1	41
16	gp_to_equity	+1	41
17	capex_to_revenue	+1	41
18	momentum_0_21	+1	41
19	debt_to_equity	+1	41
20	price_zscore_63d	+1	41
21	ma_dist_63d	+1	41

Returns

Factor	Ann. ret.	Ann. vol.	Sharpe	HAC t	Max DD	Style R2	Style alpha	Style vol.
share_turnover_21d	13.36%	5.21%	2.57	4.98	-1.42%	0.237	12.68%	5.21%
accruals_noa_proxy	5.04%	3.63%	1.39	2.72	-2.67%	0.092	5.98%	3.63%
momentum_21_252_vol_63d	8.39%	8.98%	0.93	2.23	-5.09%	0.717	6.08%	8.98%
eps_growth_yoy	4.35%	4.74%	0.92	1.58	-5.32%	0.474	3.11%	4.74%
net_income_growth_yoy	3.58%	4.11%	0.87	1.69	-6.18%	0.577	2.63%	4.11%
ret_zscore_63d	4.21%	5.33%	0.79	1.42	-7.08%	0.002	4.13%	5.33%
momentum_21_252	7.47%	9.46%	0.79	1.98	-5.50%	0.690	5.88%	9.46%
resid_ret_252v_1260c_SPY	4.54%	6.49%	0.70	1.22	-7.30%	0.023	5.40%	6.49%
dpo	3.24%	4.79%	0.68	1.28	-5.55%	0.370	5.69%	4.79%
asset_turnover	2.98%	6.05%	0.49	0.89	-11.44%	0.438	3.10%	6.05%
stoch_k_14d	0.93%	6.22%	0.15	0.31	-10.54%	0.143	1.64%	6.22%
bb_pctb_20d	0.10%	6.18%	0.02	0.04	-9.94%	0.123	0.74%	6.18%
accruals_total	0.06%	4.78%	0.01	0.02	-10.15%	0.317	2.21%	4.78%
rsi_14d	-2.34%	6.80%	-0.35	-0.73	-15.50%	0.188	-1.94%	6.80%
debt_to_assets	-1.97%	5.60%	-0.35	-0.86	-11.36%	0.572	-3.77%	5.60%
gp_to_equity	-2.86%	7.60%	-0.38	-0.63	-22.38%	0.573	-3.19%	7.60%
capex_to_revenue	-1.88%	4.53%	-0.42	-0.56	-14.04%	0.309	-0.35%	4.53%
momentum_0_21	-3.00%	6.80%	-0.44	-0.87	-20.09%	0.120	-2.76%	6.80%
debt_to_equity	-2.67%	5.20%	-0.51	-1.04	-14.16%	0.513	-4.59%	5.20%
price_zscore_63d	-3.69%	6.78%	-0.54	-1.17	-19.76%	0.451	-3.47%	6.78%
ma_dist_63d	-4.19%	7.38%	-0.57	-1.10	-21.85%	0.441	-3.49%	7.38%

Return Uncertainty

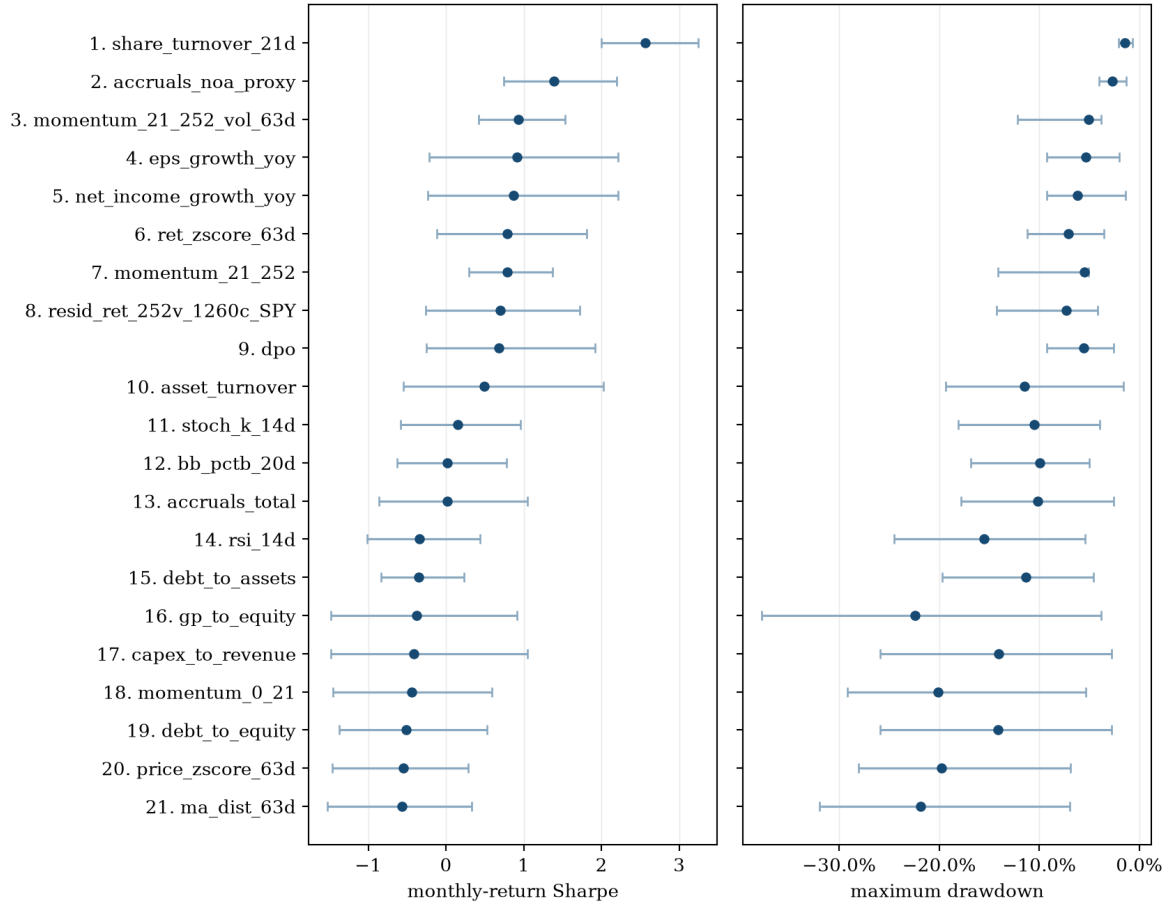


Figure: point estimates and paired stationary-bootstrap 95% intervals.

Idx	Factor	Ann. ret. [95%]	Sharpe [95%]	Pr(S<=0)	Skew [95%]	Max DD [95%]	ES 5% [95%]
1	share_turnover_21d	13.4% [9.5%, 17.1%]	2.57 [2.00, 3.25]	0.0%	0.57 [0.18, 1.11]	-1.4% [-2.1%, -0.7%]	-1.2% [-1.4%, -0.6%]
2	accruals_noa_proxy	5.0% [2.8%, 7.4%]	1.39 [0.74, 2.20]	0.0%	0.26 [-0.00, 0.67]	-2.7% [-4.0%, -1.3%]	-1.3% [-1.8%, -0.8%]
3	momentum_21_252_vol_63d	0.4% [4.0%, 13.0%]	0.93 [0.42, 1.54]	0.1%	-0.22 [-0.87, 0.68]	-5.1% [-12.2%, -3.8%]	-4.1% [-7.0%, -2.0%]
4	eps_growth_yoy	4.3% [-1.0%, 10.3%]	0.92 [-0.21, 2.22]	6.2%	0.13 [-0.30, 0.58]	-5.3% [-9.3%, -2.0%]	-2.2% [-2.8%, -1.4%]
5	net_income_growth_yoy	3.6% [-0.9%, 8.3%]	0.87 [-0.24, 2.22]	6.2%	-0.03 [-0.31, 0.25]	-6.2% [-9.3%, -1.4%]	-1.8% [-2.0%, -1.1%]
6	ret_zscore_63d	4.2% [-0.6%, 9.1%]	0.79 [-0.11, 1.81]	5.5%	-0.14 [-0.53, 0.33]	-7.1% [-11.2%, -3.5%]	-2.7% [-3.5%, -1.4%]
7	momentum_21_252	7.5% [3.0%, 11.9%]	0.79 [0.30, 1.37]	0.2%	-0.34 [-1.04, 0.75]	-5.5% [-14.1%, -5.1%]	-4.7% [-7.9%, -2.1%]
8	resid_ret_252v_1260c_SPY	1.5% [-1.7%, 10.8%]	0.70 [-0.26, 1.72]	8.1%	-0.27 [-0.72, 0.15]	-7.3% [-14.3%, -4.1%]	-3.4% [-4.6%, -2.1%]
9	dpo	3.2% [-1.2%, 8.3%]	0.68 [-0.25, 1.92]	10.0%	0.06 [-0.36, 0.55]	-5.5% [-9.3%, -2.6%]	-2.0% [-2.7%, -1.3%]
10	asset_turnover	3.0% [-3.8%, 9.2%]	0.49 [-0.55, 2.02]	18.1%	-0.55 [-0.77, 0.47]	-11.4% [-19.3%, -1.6%]	-3.6% [-3.9%, -1.4%]
11	stoch_k_14d	0.9% [-3.8%, 5.3%]	0.15 [-0.58, 0.96]	32.5%	-0.31 [-0.74, 0.39]	-10.5% [-18.1%, -3.9%]	-3.5% [-4.1%, -1.9%]
12	bb_pctb_20d	0.1% [-4.0%, 4.3%]	0.02 [-0.63, 0.78]	48.4%	-0.20 [-0.85, 0.45]	-9.9% [-16.8%, -5.0%]	-3.3% [-4.7%, -2.1%]
13	accruals_total	0.1% [-4.6%, 4.1%]	0.01 [-0.87, 1.05]	45.6%	-0.58 [-1.08, 0.31]	-10.1% [-17.8%, -2.6%]	-3.2% [-3.8%, -1.6%]
14	rsi_14d	-2.3% [-7.2%, 2.7%]	-0.35 [-1.01, 0.44]	82.8%	-0.34 [-0.73, 0.17]	-15.5% [-24.5%, -5.4%]	-3.9% [-5.4%, -2.5%]
15	debt_to_assets	-2.0% [-5.3%, 1.1%]	-0.35 [-0.84, 0.24]	88.5%	-0.92 [-1.33, -0.51]	-11.4% [-19.7%, -4.6%]	-3.9% [-4.6%, -2.8%]
16	gp_to_equity	-2.9% [-12.1%, 5.5%]	-0.38 [-1.48, 0.92]	73.4%	-0.31 [-0.65, 0.34]	-22.4% [-37.7%, -3.8%]	-4.5% [-5.4%, -2.1%]
17	capex_to_revenue	-1.9% [-7.8%, 3.3%]	-0.42 [-1.48, 1.05]	73.2%	-0.73 [-1.13, 0.23]	-14.0% [-25.9%, -2.8%]	-3.2% [-3.8%, -1.2%]
18	momentum_0_21	-3.0% [-9.2%, 3.9%]	-0.44 [-1.45, 0.59]	81.3%	0.01 [-0.30, 0.28]	-20.1% [-29.2%, -5.3%]	-4.0% [-4.2%, -2.6%]
19	debt_to_equity	-2.7% [-8.1%, 2.1%]	-0.51 [-1.37, 0.53]	84.9%	-0.56 [-1.18, 0.23]	-14.2% [-25.9%, -2.8%]	-3.6% [-4.2%, -1.8%]
20	price_zscore_63d	-3.7% [-8.8%, 2.0%]	-0.54 [-1.46, 0.28]	90.7%	0.28 [-0.39, 0.95]	-19.8% [-28.1%, -6.9%]	-3.8% [-5.6%, -2.3%]
21	ma_dist_63d	-4.2% [-10.3%, 2.6%]	-0.57 [-1.53, 0.33]	89.5%	0.25 [-0.27, 0.81]	-21.8% [-31.9%, -7.0%]	-4.1% [-5.9%, -2.8%]

Risk Loadings

Factor	MKT	SMB	VAL	MOM	BAB	QMJ	Alpha	R2
share_turnover_21d	-0.07 (-0.82)	-0.17 (-1.91)	-0.10 (-1.08)	-0.00 (-0.04)	0.05 (0.36)	0.02 (0.17)	13.90% (4.42)	0.253
accruals_noa_proxy	-0.04 (-0.50)	0.11 (2.31)	-0.02 (-0.34)	0.05 (0.88)	0.06 (0.50)	-0.02 (-0.28)	5.80% (2.14)	0.093
momentum_21_252_vol_63d	-0.20 (-2.94)	-0.11 (-1.26)	-0.28 (-3.39)	0.38 (7.72)	0.60 (3.74)	-0.22 (-1.73)	6.10% (1.79)	0.717
eps_growth_yoy	-0.06 (-0.83)	-0.10 (-1.64)	-0.25 (-2.94)	0.03 (0.52)	0.41 (2.88)	-0.14 (-1.18)	3.11% (1.35)	0.474
net_income_growth_yoy	-0.04 (-0.60)	-0.14 (-2.30)	-0.24 (-3.79)	-0.02 (-0.42)	0.35 (2.89)	-0.08 (-0.65)	2.80% (1.68)	0.578
ret_zscore_63d	-0.03 (-0.21)	-0.01 (-0.05)	0.01 (0.09)	-0.00 (-0.01)	0.01 (0.04)	-0.03 (-0.12)	4.57% (1.17)	0.004
momentum_21_252	-0.19 (-2.58)	-0.07 (-0.75)	-0.27 (-3.14)	0.42 (8.08)	0.54 (3.04)	-0.11 (-0.78)	6.00% (1.81)	0.690
resid_ret_252v_1260c_SPY	-0.06 (-0.43)	0.06 (0.34)	-0.04 (-0.24)	-0.04 (-0.38)	0.08 (0.30)	0.01 (0.05)	6.77% (1.55)	0.040
dpo	0.07 (1.25)	0.20 (4.47)	-0.16 (-2.15)	-0.15 (-3.31)	0.14 (0.98)	0.21 (2.65)	5.95% (2.72)	0.371
asset_turnover	0.30 (4.81)	0.18 (2.11)	-0.12 (-1.06)	-0.03 (-0.43)	0.05 (0.25)	0.51 (5.22)	2.72% (0.94)	0.439
stoch_k_14d	0.14 (1.49)	0.08 (0.74)	0.00 (0.03)	-0.05 (-1.27)	-0.23 (-1.17)	0.09 (0.63)	-0.85% (-0.22)	0.187
bb_pctb_20d	0.11 (1.30)	0.07 (0.70)	0.02 (0.15)	-0.07 (-1.61)	-0.17 (-0.85)	0.03 (0.24)	-1.37% (-0.44)	0.157
accruals_total	-0.02 (-0.25)	0.13 (1.39)	-0.22 (-2.91)	-0.09 (-1.30)	-0.07 (-0.50)	0.08 (0.65)	1.69% (0.64)	0.321
rsi_14d	0.14 (1.73)	0.09 (0.79)	0.04 (0.36)	-0.10 (-2.00)	-0.16 (-0.74)	-0.01 (-0.07)	-4.23% (-1.10)	0.221
debt_to_assets	0.27 (3.69)	-0.04 (-0.42)	0.08 (0.96)	-0.02 (-0.30)	0.20 (1.45)	0.32 (2.37)	-3.54% (-1.70)	0.572
gp_to_equity	0.27 (4.43)	-0.00 (-0.08)	-0.32 (-3.35)	-0.17 (-2.11)	0.42 (1.95)	0.43 (5.47)	-2.93% (-1.02)	0.573
capex_to_revenue	0.06 (0.74)	0.11 (1.56)	-0.22 (-2.85)	-0.17 (-2.55)	0.16 (0.98)	0.11 (0.82)	-0.45% (-0.13)	0.309
momentum_0_21	0.12 (1.28)	0.07 (0.42)	-0.00 (-0.02)	-0.10 (-1.21)	-0.01 (-0.03)	-0.07 (-0.62)	-5.23% (-1.13)	0.160
debt_to_equity	0.20 (3.39)	-0.09 (-1.08)	-0.03 (-0.34)	-0.08 (-1.45)	0.34 (2.54)	0.16 (1.26)	-4.34% (-2.13)	0.514
price_zscore_63d	0.15 (1.93)	0.09 (0.83)	0.03 (0.26)	-0.27 (-6.16)	-0.15 (-0.73)	-0.02 (-0.23)	-4.47% (-1.37)	0.458
ma_dist_63d	0.16 (1.72)	0.16 (1.24)	-0.01 (-0.11)	-0.28 (-6.51)	-0.13 (-0.61)	-0.01 (-0.06)	-4.76% (-1.22)	0.450

Return Contributions

Factor	MKT	SMB	VAL	MOM	BAB	QMJ	Alpha	Total
share_turnover_21d	-1.58%	0.87%	0.50%	-0.02%	-0.13%	-0.18%	13.90%	13.36%
accruals_noa_proxy	-0.90%	-0.56%	0.10%	0.49%	-0.16%	0.27%	5.80%	5.04%
momentum_21_252_vol_63d	-4.33%	0.57%	1.42%	3.81%	-1.54%	2.35%	6.10%	8.39%
eps_growth_yoy	-1.34%	0.52%	1.26%	0.29%	-1.07%	1.57%	3.11%	4.35%
net_income_growth_yoy	-0.86%	0.71%	1.24%	-0.24%	-0.89%	0.83%	2.80%	3.58%
ret_zscore_63d	-0.58%	0.04%	-0.06%	-0.01%	-0.02%	0.28%	4.57%	4.21%
momentum_21_252	-4.23%	0.36%	1.39%	4.15%	-1.39%	1.20%	6.00%	7.47%
resid_ret_252v_1260c_SPY	-1.39%	-0.31%	0.18%	-0.36%	-0.22%	-0.14%	6.77%	4.54%
dpo	1.57%	-1.01%	0.80%	-1.45%	-0.36%	-2.25%	5.95%	3.24%
asset_turnover	6.60%	-0.91%	0.60%	-0.34%	-0.13%	-5.55%	2.72%	2.98%
stoch_k_14d	3.05%	-0.40%	-0.02%	-0.49%	0.60%	-0.96%	-0.85%	0.93%
bb_pctb_20d	2.54%	-0.35%	-0.08%	-0.71%	0.43%	-0.36%	-1.37%	0.10%
accruals_total	-0.48%	-0.66%	1.13%	-0.92%	0.18%	-0.89%	1.69%	0.06%
rsi_14d	3.03%	-0.47%	-0.20%	-1.00%	0.42%	0.11%	-4.23%	-2.34%
debt_to_assets	5.98%	0.19%	-0.41%	-0.22%	-0.51%	-3.47%	-3.54%	-1.97%
gp_to_equity	5.91%	0.02%	1.62%	-1.71%	-1.08%	-4.69%	-2.93%	-2.86%
capex_to_revenue	1.28%	-0.55%	1.13%	-1.73%	-0.40%	-1.16%	-0.45%	-1.88%
momentum_0_21	2.76%	-0.33%	0.01%	-1.01%	0.02%	0.78%	-5.23%	-3.00%
debt_to_equity	4.43%	0.48%	0.15%	-0.81%	-0.86%	-1.72%	-4.34%	-2.67%
price_zscore_63d	3.42%	-0.45%	-0.14%	-2.66%	0.40%	0.22%	-4.47%	-3.69%
ma_dist_63d	3.66%	-0.79%	0.06%	-2.76%	0.33%	0.07%	-4.76%	-4.19%

1. share_turnover_21d

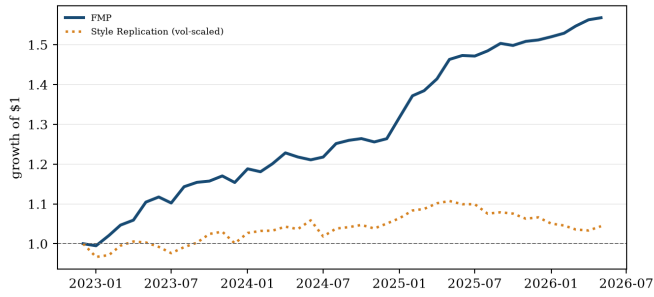


Figure: Common-window cumulative growth of \$1.

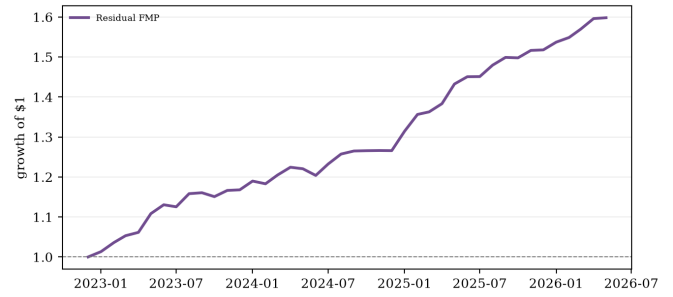


Figure: Cumulative growth after simultaneous control-factor orthogonalisation.

Factor	Ann. ret.	Ann. vol.	Sharpe	HAC t	Max DD	Style R2	Style alpha	Style vol.
share_turnover_21d	13.36%	5.21%	2.57	4.98	-1.42%	0.237	12.68%	5.21%

Factor	MKT	SMB	VAL	MOM	BAB	QMJ	Alpha	R2
share_turnover_21d	-0.07 (-0.82)	-0.17 (-1.91)	-0.10 (-1.08)	-0.00 (-0.04)	0.05 (0.36)	0.02 (0.17)	13.90% (4.42)	0.253

Factor	MKT	SMB	VAL	MOM	BAB	QMJ	Alpha	Total
share_turnover_21d	-1.58%	0.87%	0.50%	-0.02%	-0.13%	-0.18%	13.90%	13.36%

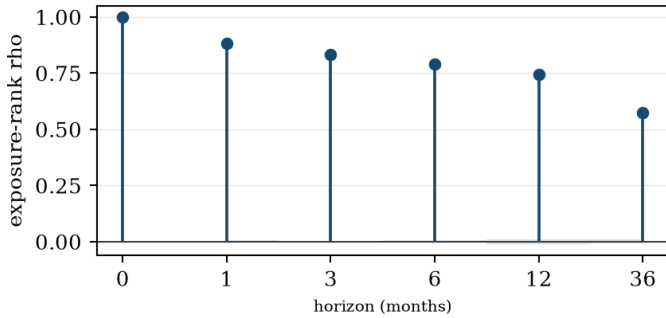


Figure: Exposure-rank ACF; 5% significant only outside the pale HAC band.

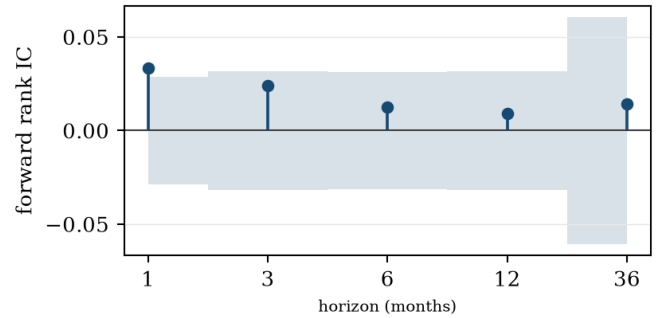


Figure: Exposure-return rank CCF; 5% significant only outside the pale HAC band.

Item	Value
Direction	+1
Common months	41
Signals	share_turnover_21d(-1)
Effective construction	rank/rank; 1%-99%; signal_weighted; LS; beta-neutral=True; group=sector; cost=0bp
Components required	1 of 1
Available / comparison months	43 / 41
Available window	2023-01-01 to 2026-07-01
Dropped: low breadth / common	0 / 2
Stocks, min / median / max	2106 / 2547 / 2812
Available-history Sharpe	2.39
Bootstrap Sharpe	2.57 [2.00, 3.25]; Pr(S<=0)=0.0%
Bootstrap tails	skew=0.57 [0.18, 1.11]; max DD=-1.4% [-2.1%, -0.7%]

2. accruals_noa_proxy

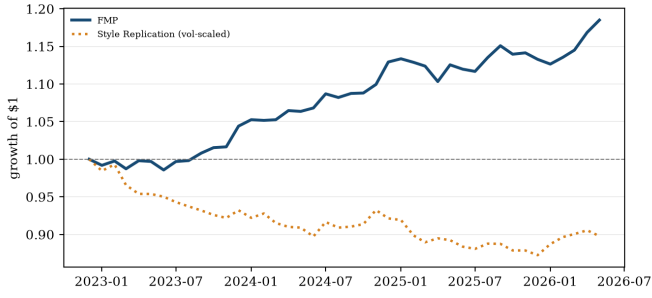


Figure: Common-window cumulative growth of \$1.

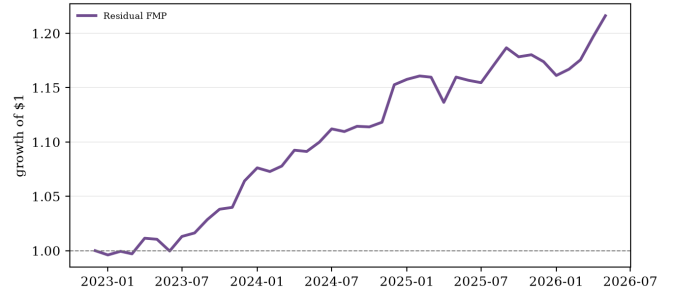


Figure: Cumulative growth after simultaneous control-factor orthogonalisation.

Factor	Ann. ret.	Ann. vol.	Sharpe	HAC t	Max DD	Style R2	Style alpha	Style vol.
accruals_noa_proxy	5.04%	3.63%	1.39	2.72	-2.67%	0.092	5.98%	3.63%

Factor	MKT	SMB	VAL	MOM	BAB	QMJ	Alpha	R2
accruals_noa_proxy	-0.04 (-0.50)	0.11 (2.31)	-0.02 (-0.34)	0.05 (0.88)	0.06 (0.50)	-0.02 (-0.28)	5.80% (2.14)	0.093

Factor	MKT	SMB	VAL	MOM	BAB	QMJ	Alpha	Total
accruals_noa_proxy	-0.90%	-0.56%	0.10%	0.49%	-0.16%	0.27%	5.80%	5.04%

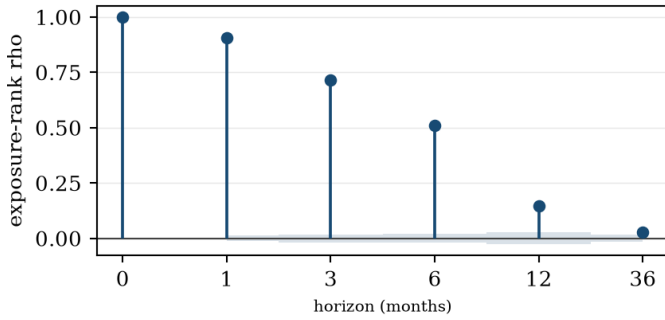


Figure: Exposure-rank ACF; 5% significant only outside the pale HAC band.

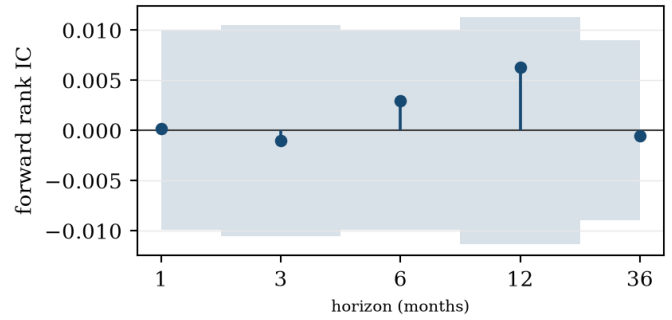


Figure: Exposure-return rank CCF; 5% significant only outside the pale HAC band.

Item	Value
Direction	+1
Common months	41
Signals	accruals_noa_proxy(-1)
Effective construction	rank/rank; 1%-99%; signal_weighted; LS; beta-neutral=True; group=sector; cost=0bp
Components required	1 of 1
Available / comparison months	43 / 41
Available window	2023-01-01 to 2026-07-01
Dropped: low breadth / common	0 / 2
Stocks, min / median / max	2034 / 2475 / 2685
Available-history Sharpe	1.38
Bootstrap Sharpe	1.39 [0.74, 2.20]; Pr(S<=0)=0.0%
Bootstrap tails	skew=0.26 [-0.00, 0.67]; max DD=-2.7% [-4.0%, -1.3%]

3. momentum_21_252_vol_63d

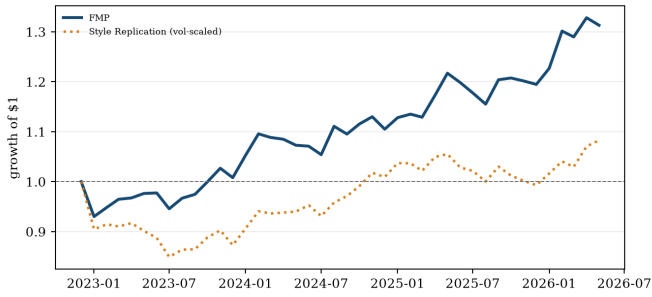


Figure: Common-window cumulative growth of \$1.

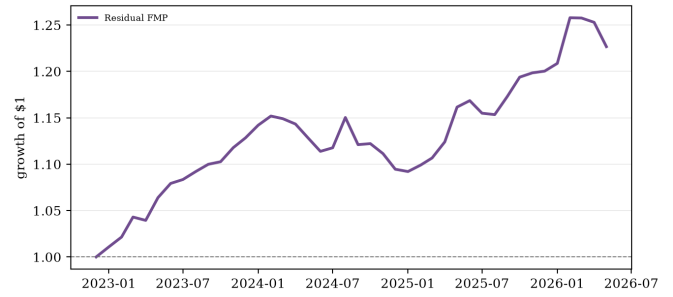


Figure: Cumulative growth after simultaneous control-factor orthogonalisation.

Factor	Ann. ret.	Ann. vol.	Sharpe	HAC t	Max DD	Style R2	Style alpha	Style vol.
momentum_21_252_vol_63d	8.39%	8.98%	0.93	2.23	-5.09%	0.717	6.08%	8.98%

Factor	MKT	SMB	VAL	MOM	BAB	QMJ	Alpha	R2
momentum_21_252_vol_63d	-0.20	-0.11	-0.28	0.38	0.60	-0.22	6.10%	0.717
	(-2.94)	(-1.26)	(-3.39)	(7.72)	(3.74)	(-1.73)	(1.79)	

Factor	MKT	SMB	VAL	MOM	BAB	QMJ	Alpha	Total
momentum_21_252_vol_63d	-4.33%	0.57%	1.42%	3.81%	-1.54%	2.35%	6.10%	8.39%

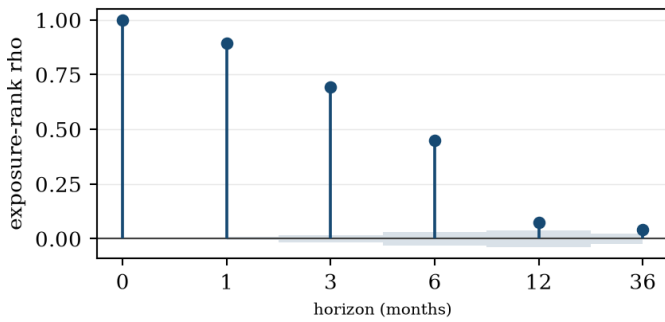


Figure: Exposure-rank ACF; 5% significant only outside the pale HAC band.

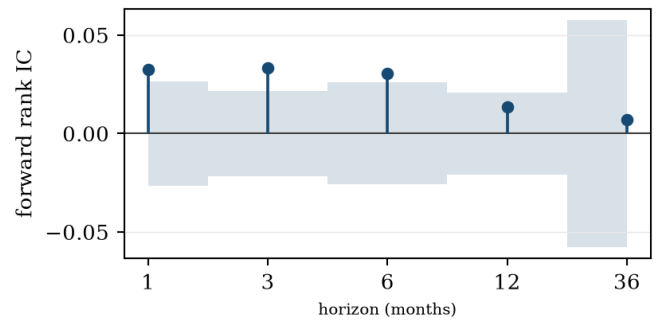


Figure: Exposure-return rank CCF; 5% significant only outside the pale HAC band.

Item	Value
Direction	+1
Common months	41
Signals	momentum_21_252_vol_63d(+1)
Effective construction	rank/rank; 1%-99%; signal_weighted; LS; beta-neutral=True; group=sector; cost=0bp
Components required	1 of 1
Available / comparison months	43 / 41
Available window	2023-01-01 to 2026-07-01
Dropped: low breadth / common	0 / 2
Stocks, min / median / max	2106 / 2547 / 2811
Available-history Sharpe	0.66
Bootstrap Sharpe	0.93 [0.42, 1.54]; Pr(S<=0)=0.1%
Bootstrap tails	skew=-0.22 [-0.87, 0.68]; max DD=-5.1% [-12.2%, -3.8%]

4. eps_growth_yoy

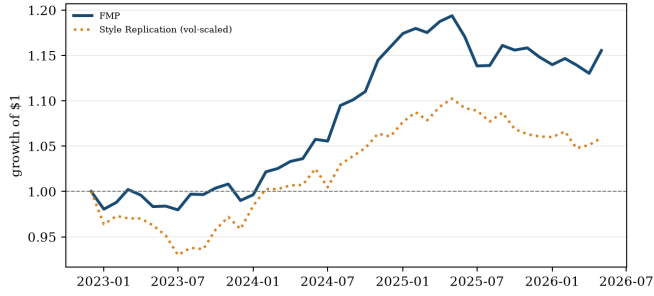


Figure: Common-window cumulative growth of \$1.

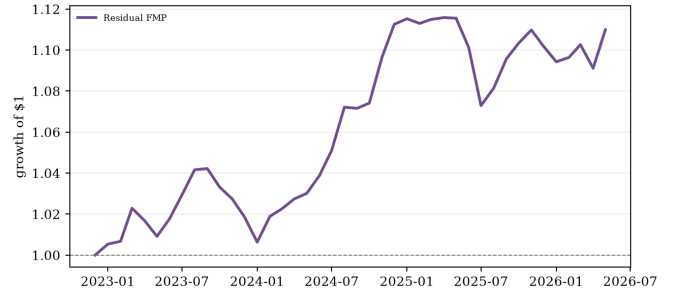


Figure: Cumulative growth after simultaneous control-factor orthogonalisation.

Factor	Ann. ret.	Ann. vol.	Sharpe	HAC t	Max DD	Style R2	Style alpha	Style vol.
eps_growth_yoy	4.35%	4.74%	0.92	1.58	-5.32%	0.474	3.11%	4.74%

Factor	MKT	SMB	VAL	MOM	BAB	QMJ	Alpha	R2
eps_growth_yoy	-0.06 (-0.83)	-0.10 (-1.64)	-0.25 (-2.94)	0.03 (0.52)	0.41 (2.88)	-0.14 (-1.18)	3.11% (1.35)	0.474

Factor	MKT	SMB	VAL	MOM	BAB	QMJ	Alpha	Total
eps_growth_yoy	-1.34%	0.52%	1.26%	0.29%	-1.07%	1.57%	3.11%	4.35%

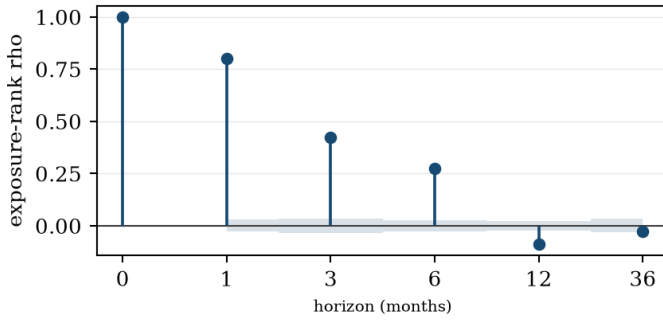


Figure: Exposure-rank ACF; 5% significant only outside the pale HAC band.

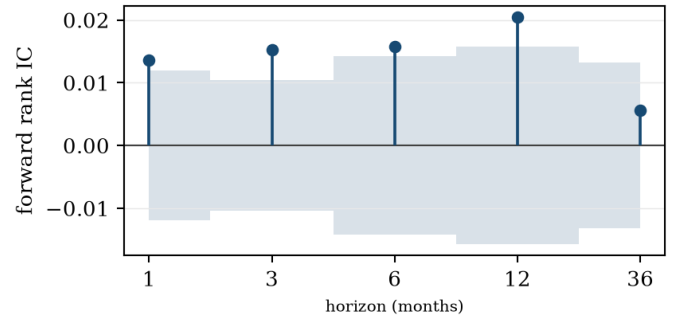


Figure: Exposure-return rank CCF; 5% significant only outside the pale HAC band.

Item	Value
Direction	+1
Common months	41
Signals	eps_growth_yoy(+1)
Effective construction	rank/rank; 1%-99%; signal_weighted; LS; beta-neutral=True; group=sector; cost=0bp
Components required	1 of 1
Available / comparison months	43 / 41
Available window	2023-01-01 to 2026-07-01
Dropped: low breadth / common	0 / 2
Stocks, min / median / max	2061 / 2479 / 2709
Available-history Sharpe	0.93
Bootstrap Sharpe	0.92 [-0.21, 2.22]; Pr(S<=0)=6.2%
Bootstrap tails	skew=0.13 [-0.30, 0.58]; max DD=-5.3% [-9.3%, -2.0%]

5. net_income_growth_yoy

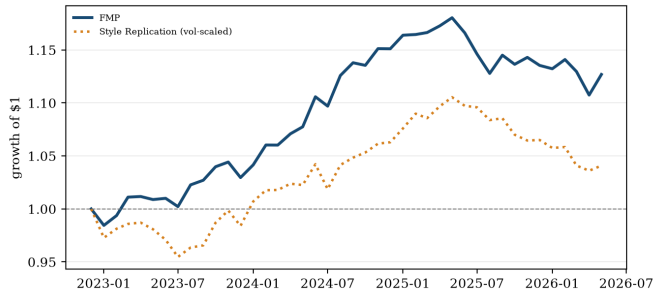


Figure: Common-window cumulative growth of \$1.

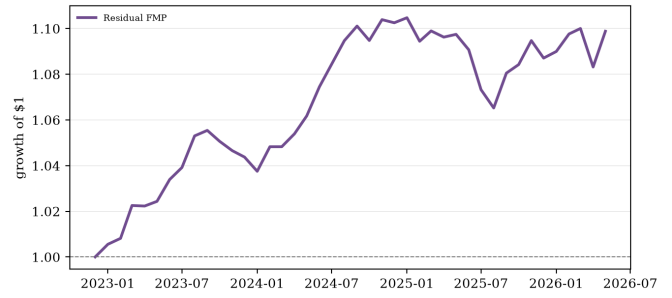


Figure: Cumulative growth after simultaneous control-factor orthogonalisation.

Factor	Ann. ret.	Ann. vol.	Sharpe	HAC t	Max DD	Style R2	Style alpha	Style vol.
net_income_growth_yoy	3.58%	4.11%	0.87	1.69	-6.18%	0.577	2.63%	4.11%

Factor	MKT	SMB	VAL	MOM	BAB	QMJ	Alpha	R2
net_income_growth_yoy	-0.04	-0.14	-0.24	-0.02	0.35	-0.08	2.80%	0.578
	(-0.60)	(-2.30)	(-3.79)	(-0.42)	(2.89)	(-0.65)	(1.68)	

Factor	MKT	SMB	VAL	MOM	BAB	QMJ	Alpha	Total
net_income_growth_yoy	-0.86%	0.71%	1.24%	-0.24%	-0.89%	0.83%	2.80%	3.58%

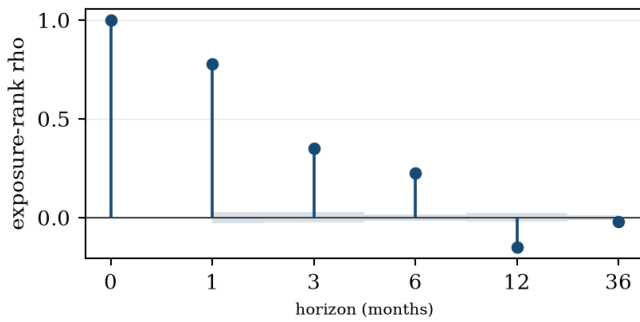


Figure: Exposure-rank ACF; 5% significant only outside the pale HAC band.

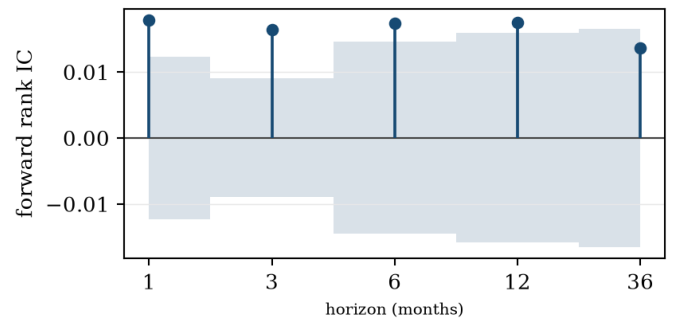


Figure: Exposure-return rank CCF; 5% significant only outside the pale HAC band.

Item	Value
Direction	+1
Common months	41
Signals	net_income_growth_yoy(+1)
Effective construction	rank/rank; 1%-99%; signal_weighted; LS; beta-neutral=True; group=sector; cost=0bp
Components required	1 of 1
Available / comparison months	43 / 41
Available window	2023-01-01 to 2026-07-01
Dropped: low breadth / common	0 / 2
Stocks, min / median / max	2088 / 2531 / 2779
Available-history Sharpe	1.03
Bootstrap Sharpe	0.87 [-0.24, 2.22]; Pr(S<=0)=6.2%
Bootstrap tails	skew=-0.03 [-0.31, 0.25]; max DD=-6.2% [-9.3%, -1.4%]

6. ret_zscore_63d

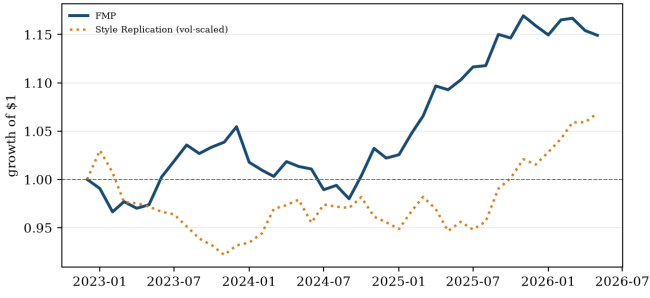


Figure: Common-window cumulative growth of \$1.

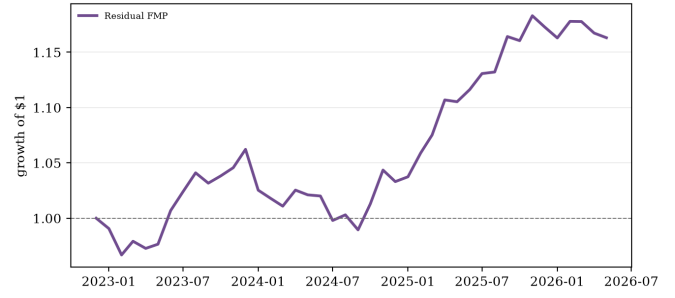


Figure: Cumulative growth after simultaneous control-factor orthogonalisation.

Factor	Ann. ret.	Ann. vol.	Sharpe	HAC t	Max DD	Style R2	Style alpha	Style vol.
ret_zscore_63d	4.21%	5.33%	0.79	1.42	-7.08%	0.002	4.13%	5.33%

Factor	MKT	SMB	VAL	MOM	BAB	QMJ	Alpha	R2
ret_zscore_63d	-0.03 (-0.21)	-0.01 (-0.05)	0.01 (0.09)	-0.00 (-0.01)	0.01 (0.04)	-0.03 (-0.12)	4.57% (1.17)	0.004

Factor	MKT	SMB	VAL	MOM	BAB	QMJ	Alpha	Total
ret_zscore_63d	-0.58%	0.04%	-0.06%	-0.01%	-0.02%	0.28%	4.57%	4.21%

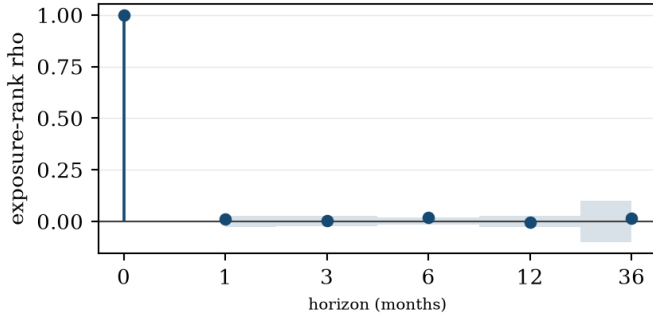


Figure: Exposure-rank ACF; 5% significant only outside the pale HAC band.

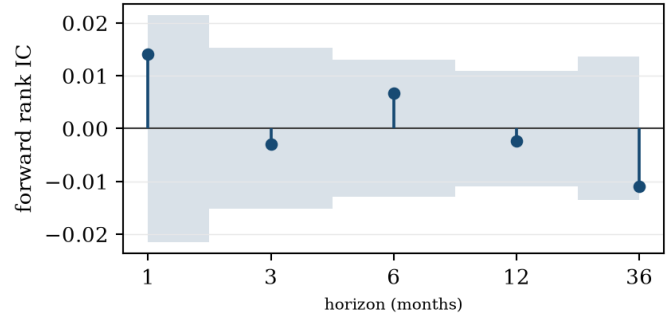


Figure: Exposure-return rank CCF; 5% significant only outside the pale HAC band.

Item	Value
Direction	+1
Common months	41
Signals	ret_zscore_63d(-1)
Effective construction	rank/rank; 1%-99%; signal_weighted; LS; beta-neutral=True; group=sector; cost=0bp
Components required	1 of 1
Available / comparison months	43 / 41
Available window	2023-01-01 to 2026-07-01
Dropped: low breadth / common	0 / 2
Stocks, min / median / max	2106 / 2547 / 2811
Available-history Sharpe	1.02
Bootstrap Sharpe	0.79 [-0.11, 1.81]; Pr(S<=0)=5.5%
Bootstrap tails	skew=-0.14 [-0.53, 0.33]; max DD=-7.1% [-11.2%, -3.5%]

7. momentum_21_252

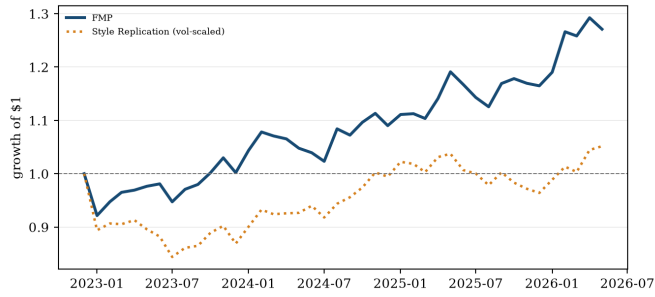


Figure: Common-window cumulative growth of \$1.

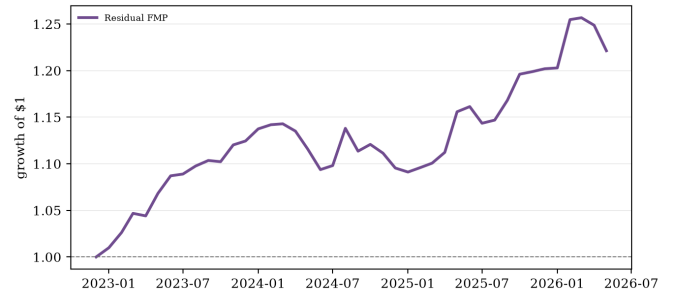


Figure: Cumulative growth after simultaneous control-factor orthogonalisation.

Factor	Ann. ret.	Ann. vol.	Sharpe	HAC t	Max DD	Style R2	Style alpha	Style vol.
momentum_21_252	7.47%	9.46%	0.79	1.98	-5.50%	0.690	5.88%	9.46%

Factor	MKT	SMB	VAL	MOM	BAB	QMJ	Alpha	R2
momentum_21_252	-0.19 (-2.58)	-0.07 (-0.75)	-0.27 (-3.14)	0.42 (8.08)	0.54 (3.04)	-0.11 (-0.78)	6.00% (1.81)	0.690

Factor	MKT	SMB	VAL	MOM	BAB	QMJ	Alpha	Total
momentum_21_252	-4.23%	0.36%	1.39%	4.15%	-1.39%	1.20%	6.00%	7.47%

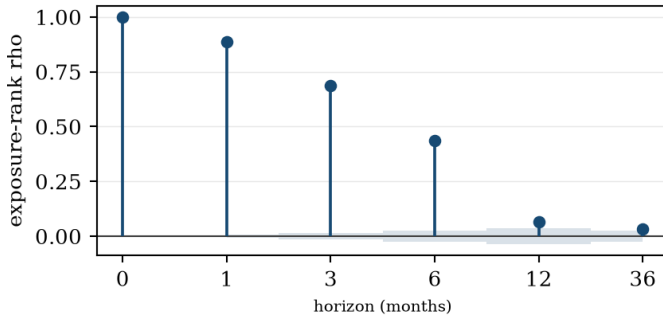


Figure: Exposure-rank ACF; 5% significant only outside the pale HAC band.

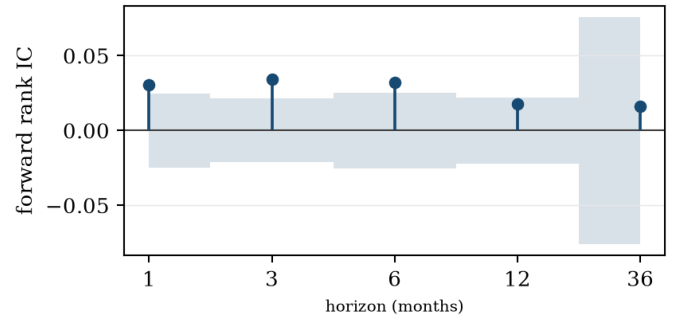


Figure: Exposure-return rank CCF; 5% significant only outside the pale HAC band.

Item	Value
Direction	+1
Common months	41
Signals	momentum_21_252(+1)
Effective construction	rank/rank; 1%-99%; signal_weighted; LS; beta-neutral=True; group=sector; cost=0bp
Components required	1 of 1
Available / comparison months	43 / 41
Available window	2023-01-01 to 2026-07-01
Dropped: low breadth / common	0 / 2
Stocks, min / median / max	2106 / 2547 / 2812
Available-history Sharpe	0.53
Bootstrap Sharpe	0.79 [0.30, 1.37]; Pr(S<=0)=0.2%
Bootstrap tails	skew=-0.34 [-1.04, 0.75]; max DD=-5.5% [-14.1%, -5.1%]

8. resid_ret_252v_1260c_SPY

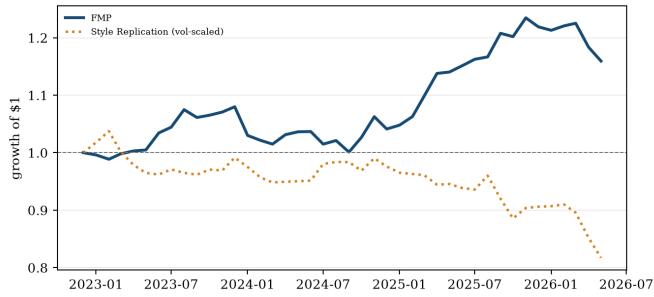


Figure: Common-window cumulative growth of \$1.

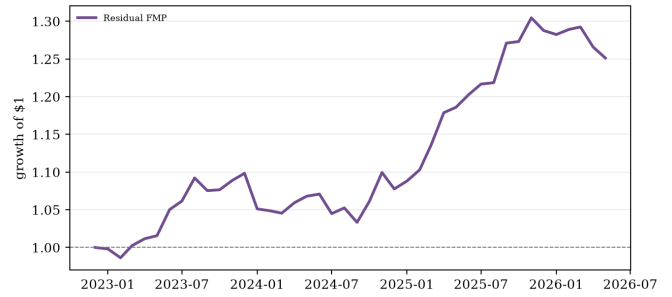


Figure: Cumulative growth after simultaneous control-factor orthogonalisation.

Factor	Ann. ret.	Ann. vol.	Sharpe	HAC t	Max DD	Style R2	Style alpha	Style vol.
resid_ret_252v_1260c_SPY	4.54%	6.49%	0.70	1.22	-7.30%	0.023	5.40%	6.49%

Factor	MKT	SMB	VAL	MOM	BAB	QMJ	Alpha	R2
resid_ret_252v_1260c_SPY	-0.06	0.06	-0.04	-0.04	0.08	0.01	6.77%	0.040
	(-0.43)	(0.34)	(-0.24)	(-0.38)	(0.30)	(0.05)	(1.55)	

Factor	MKT	SMB	VAL	MOM	BAB	QMJ	Alpha	Total
resid_ret_252v_1260c_SPY	-1.39%	-0.31%	0.18%	-0.36%	-0.22%	-0.14%	6.77%	4.54%

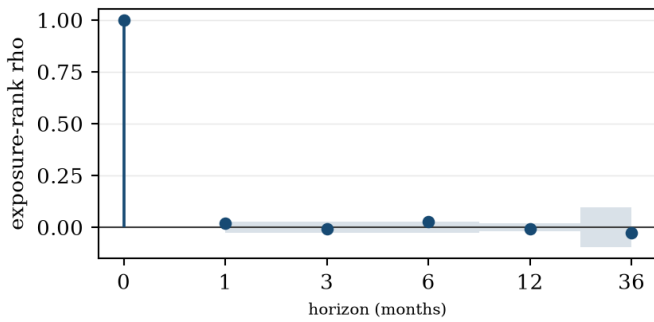


Figure: Exposure-rank ACF; 5% significant only outside the pale HAC band.

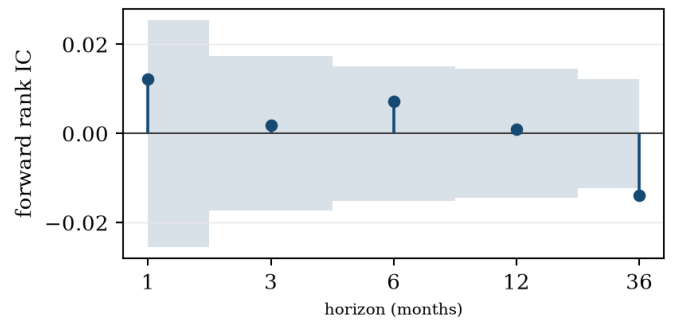


Figure: Exposure-return rank CCF; 5% significant only outside the pale HAC band.

Item	Value
Direction	+1
Common months	41
Signals	resid_ret_252v_1260c_SPY(-1)
Effective construction	rank/rank; 1%-99%; signal_weighted; LS; beta-neutral=True; group=sector; cost=0bp
Components required	1 of 1
Available / comparison months	43 / 41
Available window	2023-01-01 to 2026-07-01
Dropped: low breadth / common	0 / 2
Stocks, min / median / max	2106 / 2547 / 2812
Available-history Sharpe	0.92
Bootstrap Sharpe	0.70 [-0.26, 1.72]; Pr(S<=0)=8.1%
Bootstrap tails	skew=-0.27 [-0.72, 0.15]; max DD=-7.3% [-14.3%, -4.1%]

9. dpo

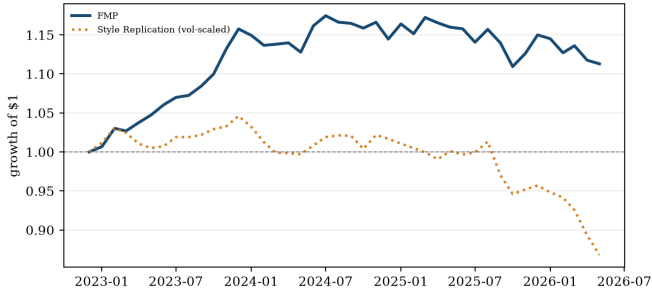


Figure: Common-window cumulative growth of \$1.

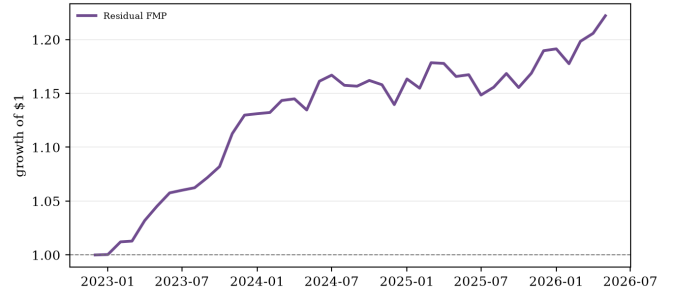


Figure: Cumulative growth after simultaneous control-factor orthogonalisation.

Factor	Ann. ret.	Ann. vol.	Sharpe	HAC t	Max DD	Style R2	Style alpha	Style vol.
dpo	3.24%	4.79%	0.68	1.28	-5.55%	0.370	5.69%	4.79%

Factor	MKT	SMB	VAL	MOM	BAB	QMJ	Alpha	R2
dpo	0.07 (1.25)	0.20 (4.47)	-0.16 (-2.15)	-0.15 (-3.31)	0.14 (0.98)	0.21 (2.65)	5.95% (2.72)	0.371

Factor	MKT	SMB	VAL	MOM	BAB	QMJ	Alpha	Total
dpo	1.57%	-1.01%	0.80%	-1.45%	-0.36%	-2.25%	5.95%	3.24%

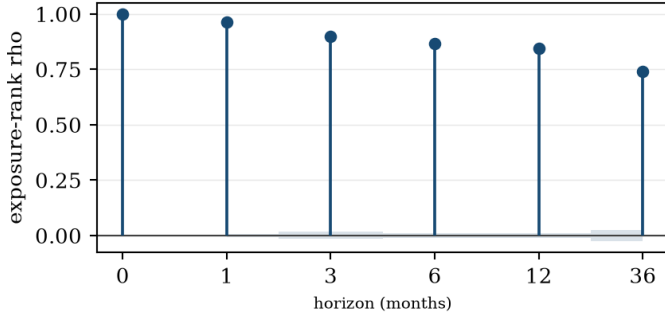


Figure: Exposure-rank ACF; 5% significant only outside the pale HAC band.

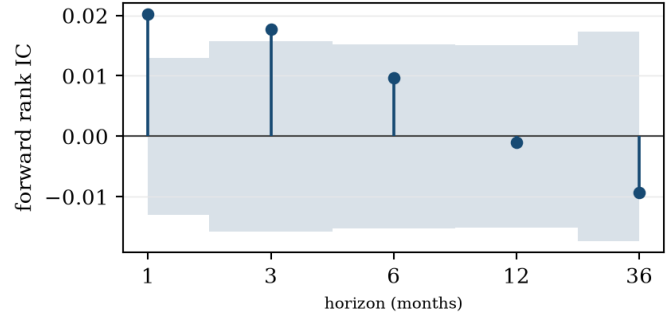


Figure: Exposure-return rank CCF; 5% significant only outside the pale HAC band.

Item	Value
Direction	+1
Common months	41
Signals	dpo(-1)
Effective construction	rank/rank; 1%-99%; signal_weighted; LS; beta-neutral=True; group=sector; cost=0bp
Components required	1 of 1
Available / comparison months	43 / 41
Available window	2023-01-01 to 2026-07-01
Dropped: low breadth / common	0 / 2
Stocks, min / median / max	2017 / 2422 / 2524
Available-history Sharpe	0.96
Bootstrap Sharpe	0.68 [-0.25, 1.92]; Pr(S<=0)=10.0%
Bootstrap tails	skew=0.06 [-0.36, 0.55]; max DD=-5.5% [-9.3%, -2.6%]

10. asset_turnover

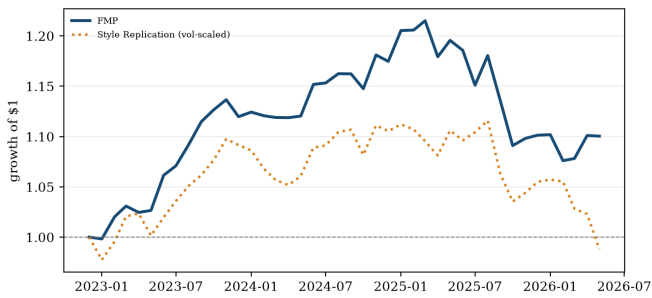


Figure: Common-window cumulative growth of \$1.

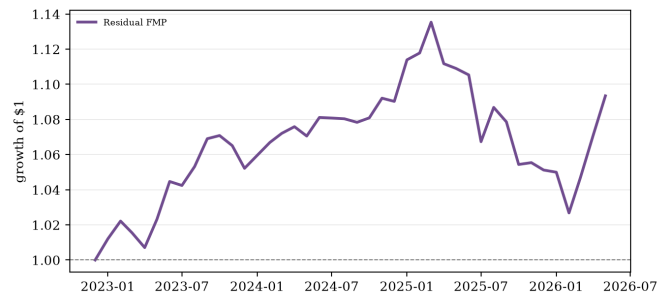


Figure: Cumulative growth after simultaneous control-factor orthogonalisation.

Factor	Ann. ret.	Ann. vol.	Sharpe	HAC t	Max DD	Style R2	Style alpha	Style vol.
asset_turnover	2.98%	6.05%	0.49	0.89	-11.44%	0.438	3.10%	6.05%

Factor	MKT	SMB	VAL	MOM	BAB	QMJ	Alpha	R2
asset_turnover	0.30 (4.81)	0.18 (2.11)	-0.12 (-1.06)	-0.03 (-0.43)	0.05 (0.25)	0.51 (5.22)	2.72% (0.94)	0.439

Factor	MKT	SMB	VAL	MOM	BAB	QMJ	Alpha	Total
asset_turnover	6.60%	-0.91%	0.60%	-0.34%	-0.13%	-5.55%	2.72%	2.98%

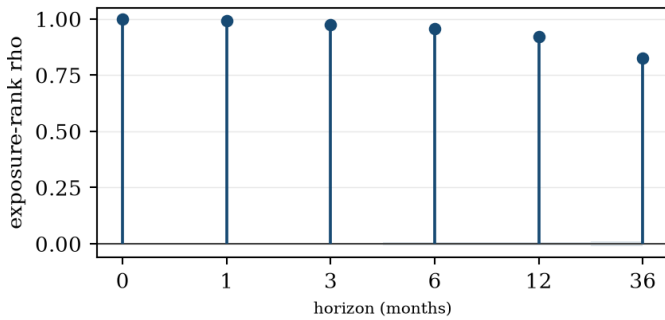


Figure: Exposure-rank ACF; 5% significant only outside the pale HAC band.

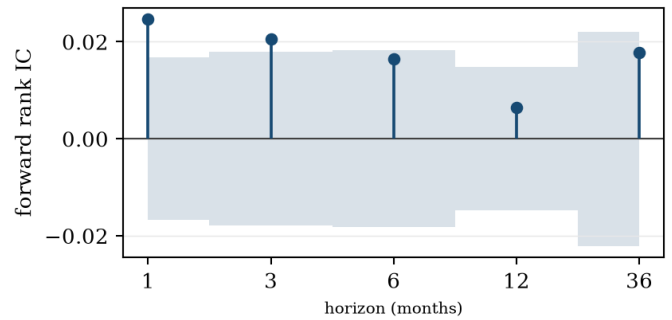


Figure: Exposure-return rank CCF; 5% significant only outside the pale HAC band.

Item	Value
Direction	+1
Common months	41
Signals	asset_turnover(+1)
Effective construction	rank/rank; 1%-99%; signal_weighted; LS; beta-neutral=True; group=sector; cost=0bp
Components required	1 of 1
Available / comparison months	43 / 41
Available window	2023-01-01 to 2026-07-01
Dropped: low breadth / common	0 / 2
Stocks, min / median / max	2086 / 2500 / 2727
Available-history Sharpe	0.81
Bootstrap Sharpe	0.49 [-0.55, 2.02]; Pr(S<=0)=18.1%
Bootstrap tails	skew=-0.55 [-0.77, 0.47]; max DD=-11.4% [-19.3%, -1.6%]

11. stoch_k_14d

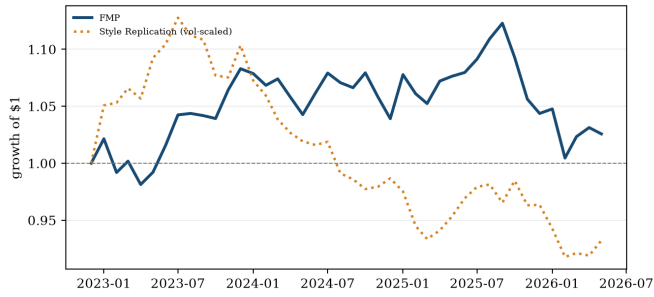


Figure: Common-window cumulative growth of \$1.

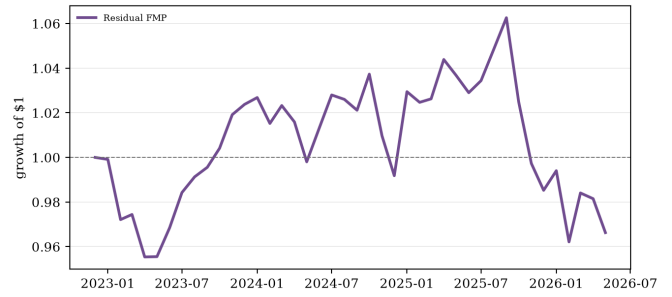


Figure: Cumulative growth after simultaneous control-factor orthogonalisation.

Factor	Ann. ret.	Ann. vol.	Sharpe	HAC t	Max DD	Style R2	Style alpha	Style vol.
stoch_k_14d	0.93%	6.22%	0.15	0.31	-10.54%	0.143	1.64%	6.22%

Factor	MKT	SMB	VAL	MOM	BAB	QMJ	Alpha	R2
stoch_k_14d	0.14 (1.49)	0.08 (0.74)	0.00 (0.03)	-0.05 (-1.27)	-0.23 (-1.17)	0.09 (0.63)	-0.85% (-0.22)	0.187

Factor	MKT	SMB	VAL	MOM	BAB	QMJ	Alpha	Total
stoch_k_14d	3.05%	-0.40%	-0.02%	-0.49%	0.60%	-0.96%	-0.85%	0.93%

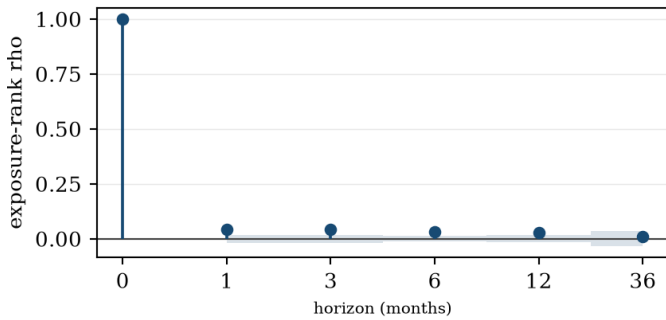


Figure: Exposure-rank ACF; 5% significant only outside the pale HAC band.

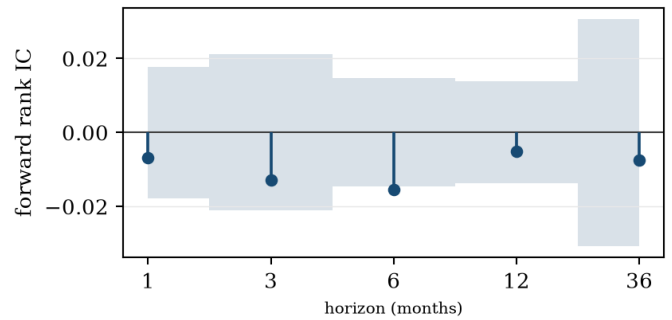


Figure: Exposure-return rank CCF; 5% significant only outside the pale HAC band.

Item	Value
Direction	+1
Common months	41
Signals	stoch_k_14d(-1)
Effective construction	rank/rank; 1%-99%; signal_weighted; LS; beta-neutral=True; group=sector; cost=0bp
Components required	1 of 1
Available / comparison months	43 / 41
Available window	2023-01-01 to 2026-07-01
Dropped: low breadth / common	0 / 2
Stocks, min / median / max	2105 / 2546 / 2810
Available-history Sharpe	0.32
Bootstrap Sharpe	0.15 [-0.58, 0.96]; Pr(S<=0)=32.5%
Bootstrap tails	skew=-0.31 [-0.74, 0.39]; max DD=-10.5% [-18.1%, -3.9%]

12. bb_pctb_20d

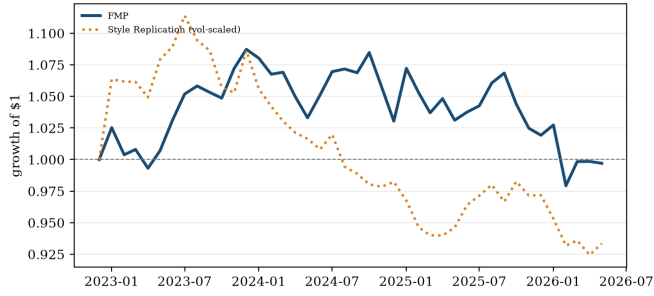


Figure: Common-window cumulative growth of \$1.

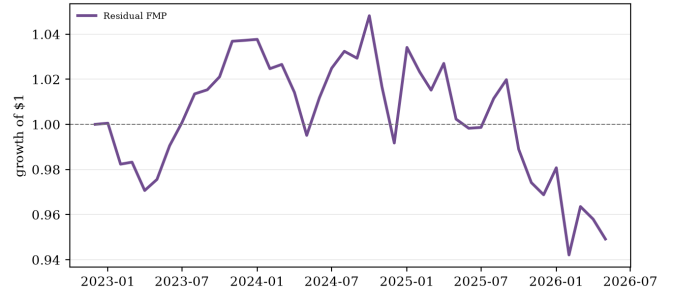


Figure: Cumulative growth after simultaneous control-factor orthogonalisation.

Factor	Ann. ret.	Ann. vol.	Sharpe	HAC t	Max DD	Style R2	Style alpha	Style vol.
bb_pctb_20d	0.10%	6.18%	0.02	0.04	-9.94%	0.123	0.74%	6.18%

Factor	MKT	SMB	VAL	MOM	BAB	QMJ	Alpha	R2
bb_pctb_20d	0.11 (1.30)	0.07 (0.70)	0.02 (0.15)	-0.07 (-1.61)	-0.17 (-0.85)	0.03 (0.24)	-1.37% (-0.44)	0.157

Factor	MKT	SMB	VAL	MOM	BAB	QMJ	Alpha	Total
bb_pctb_20d	2.54%	-0.35%	-0.08%	-0.71%	0.43%	-0.36%	-1.37%	0.10%

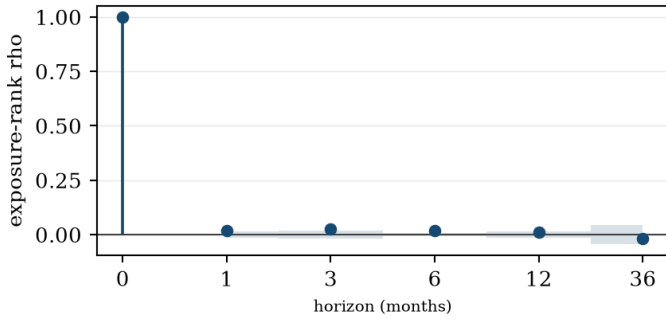


Figure: Exposure-rank ACF; 5% significant only outside the pale HAC band.

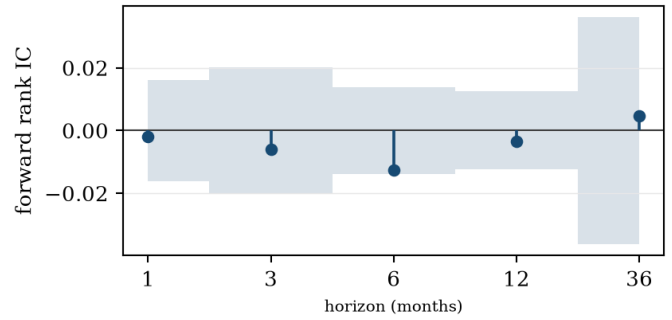


Figure: Exposure-return rank CCF; 5% significant only outside the pale HAC band.

Item	Value
Direction	+1
Common months	41
Signals	bb_pctb_20d(-1)
Effective construction	rank/rank; 1%-99%; signal_weighted; LS; beta-neutral=True; group=sector; cost=0bp
Components required	1 of 1
Available / comparison months	43 / 41
Available window	2023-01-01 to 2026-07-01
Dropped: low breadth / common	0 / 2
Stocks, min / median / max	2105 / 2546 / 2811
Available-history Sharpe	0.26
Bootstrap Sharpe	0.02 [-0.63, 0.78]; Pr(S<=0)=48.4%
Bootstrap tails	skew=-0.20 [-0.85, 0.45]; max DD=-9.9% [-16.8%, -5.0%]

13. accruals_total

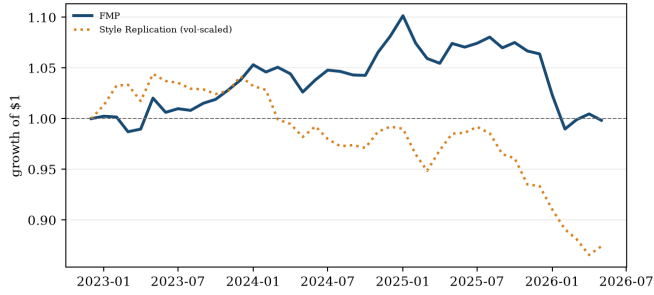


Figure: Common-window cumulative growth of \$1.

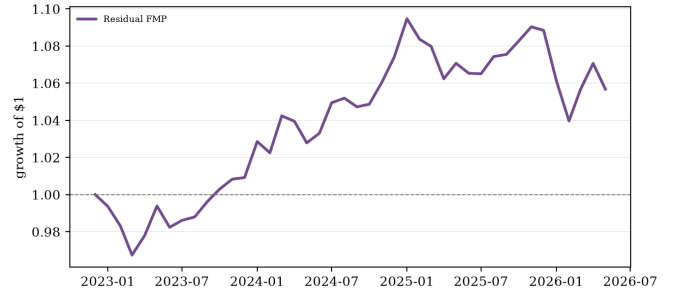


Figure: Cumulative growth after simultaneous control-factor orthogonalisation.

Factor	Ann. ret.	Ann. vol.	Sharpe	HAC t	Max DD	Style R2	Style alpha	Style vol.
accruals_total	0.06%	4.78%	0.01	0.02	-10.15%	0.317	2.21%	4.78%

Factor	MKT	SMB	VAL	MOM	BAB	QMJ	Alpha	R2
accruals_total	-0.02 (-0.25)	0.13 (1.39)	-0.22 (-2.91)	-0.09 (-1.30)	-0.07 (-0.50)	0.08 (0.65)	1.69% (0.64)	0.321

Factor	MKT	SMB	VAL	MOM	BAB	QMJ	Alpha	Total
accruals_total	-0.48%	-0.66%	1.13%	-0.92%	0.18%	-0.89%	1.69%	0.06%

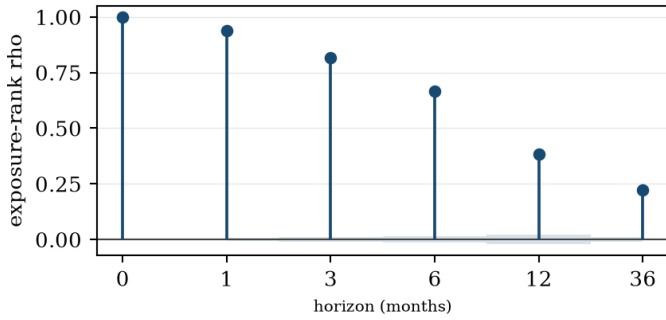


Figure: Exposure-rank ACF; 5% significant only outside the pale HAC band.

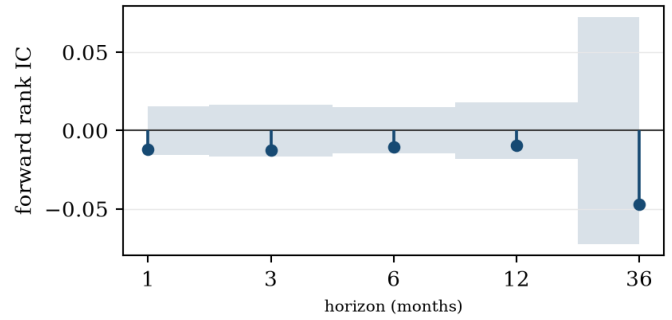


Figure: Exposure-return rank CCF; 5% significant only outside the pale HAC band.

Item	Value
Direction	+1
Common months	41
Signals	accruals_total(-1)
Effective construction	rank/rank; 1%-99%; signal_weighted; LS; beta-neutral=True; group=sector; cost=0bp
Components required	1 of 1
Available / comparison months	43 / 41
Available window	2023-01-01 to 2026-07-01
Dropped: low breadth / common	0 / 2
Stocks, min / median / max	2059 / 2454 / 2672
Available-history Sharpe	0.16
Bootstrap Sharpe	0.01 [-0.87, 1.05]; Pr(S<=0)=45.6%
Bootstrap tails	skew=-0.58 [-1.08, 0.31]; max DD=-10.1% [-17.8%, -2.6%]

14. rsi_14d

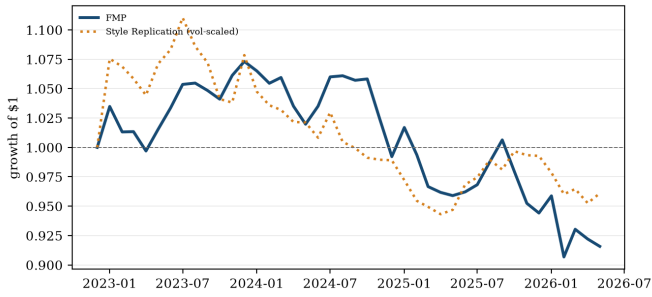


Figure: Common-window cumulative growth of \$1.

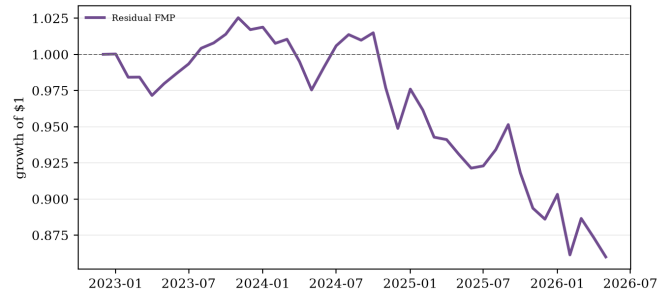


Figure: Cumulative growth after simultaneous control-factor orthogonalisation.

Factor	Ann. ret.	Ann. vol.	Sharpe	HAC t	Max DD	Style R2	Style alpha	Style vol.
rsi_14d	-2.34%	6.80%	-0.35	-0.73	-15.50%	0.188	-1.94%	6.80%

Factor	MKT	SMB	VAL	MOM	BAB	QMJ	Alpha	R2
rsi_14d	0.14 (1.73)	0.09 (0.79)	0.04 (0.36)	-0.10 (-2.00)	-0.16 (-0.74)	-0.01 (-0.07)	-4.23% (-1.10)	0.221

Factor	MKT	SMB	VAL	MOM	BAB	QMJ	Alpha	Total
rsi_14d	3.03%	-0.47%	-0.20%	-1.00%	0.42%	0.11%	-4.23%	-2.34%

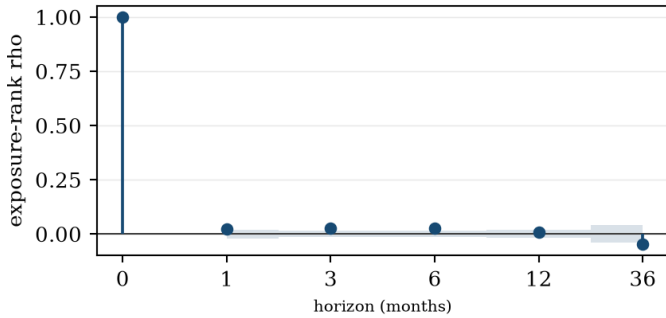


Figure: Exposure-rank ACF; 5% significant only outside the pale HAC band.

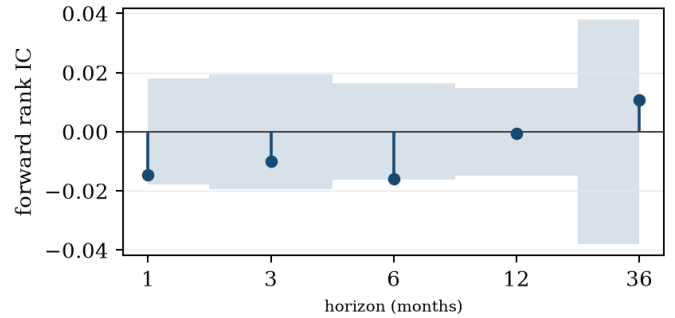


Figure: Exposure-return rank CCF; 5% significant only outside the pale HAC band.

Item	Value
Direction	+1
Common months	41
Signals	rsi_14d(-1)
Effective construction	rank/rank; 1%-99%; signal_weighted; LS; beta-neutral=True; group=sector; cost=0bp
Components required	1 of 1
Available / comparison months	43 / 41
Available window	2023-01-01 to 2026-07-01
Dropped: low breadth / common	0 / 2
Stocks, min / median / max	2105 / 2546 / 2810
Available-history Sharpe	-0.01
Bootstrap Sharpe	-0.35 [-1.01, 0.44]; Pr(S<=0)=82.8%
Bootstrap tails	skew=-0.34 [-0.73, 0.17]; max DD=-15.5% [-24.5%, -5.4%]

15. debt_to_assets

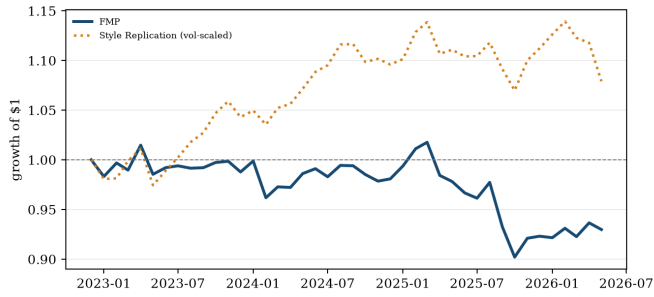


Figure: Common-window cumulative growth of \$1.

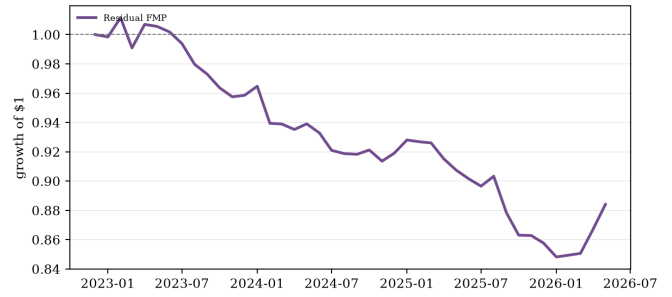


Figure: Cumulative growth after simultaneous control-factor orthogonalisation.

Factor	Ann. ret.	Ann. vol.	Sharpe	HAC t	Max DD	Style R2	Style alpha	Style vol.
debt_to_assets	-1.97%	5.60%	-0.35	-0.86	-11.36%	0.572	-3.77%	5.60%

Factor	MKT	SMB	VAL	MOM	BAB	QMJ	Alpha	R2
debt_to_assets	0.27 (3.69)	-0.04 (-0.42)	0.08 (0.96)	-0.02 (-0.30)	0.20 (1.45)	0.32 (2.37)	-3.54% (-1.70)	0.572

Factor	MKT	SMB	VAL	MOM	BAB	QMJ	Alpha	Total
debt_to_assets	5.98%	0.19%	-0.41%	-0.22%	-0.51%	-3.47%	-3.54%	-1.97%

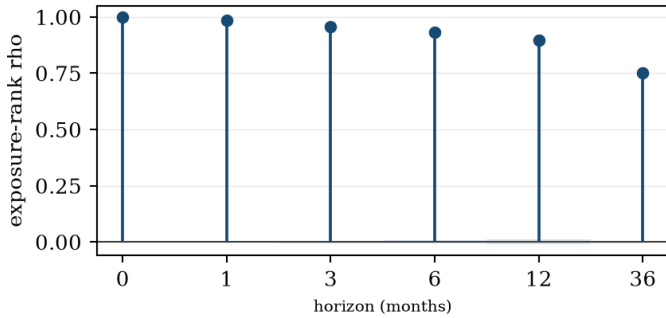


Figure: Exposure-rank ACF; 5% significant only outside the pale HAC band.

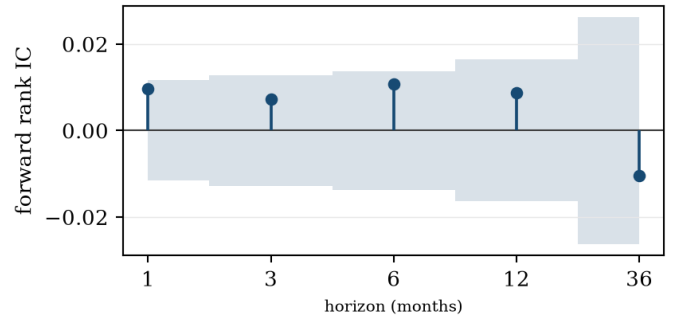


Figure: Exposure-return rank CCF; 5% significant only outside the pale HAC band.

Item	Value
Direction	+1
Common months	41
Signals	debt_to_assets(+1)
Effective construction	rank/rank; 1%-99%; signal_weighted; LS; beta-neutral=True; group=sector; cost=0bp
Components required	1 of 1
Available / comparison months	43 / 41
Available window	2023-01-01 to 2026-07-01
Dropped: low breadth / common	0 / 2
Stocks, min / median / max	2063 / 2502 / 2731
Available-history Sharpe	-0.12
Bootstrap Sharpe	-0.35 [-0.84, 0.24]; Pr(S<=0)=88.5%
Bootstrap tails	skew=-0.92 [-1.33, -0.51]; max DD=-11.4% [-19.7%, -4.6%]

16. gp_to_equity

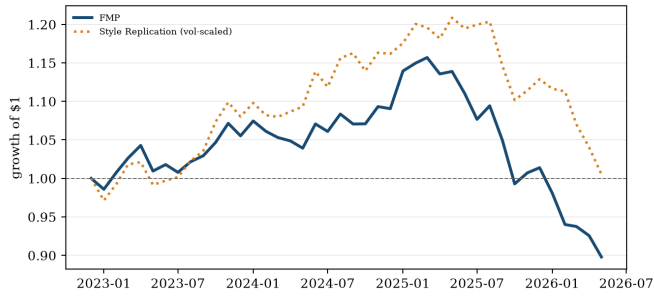


Figure: Common-window cumulative growth of \$1.

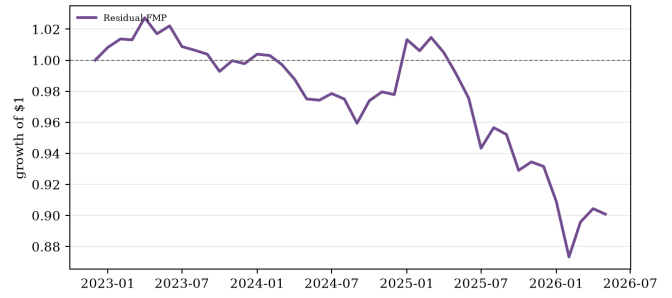


Figure: Cumulative growth after simultaneous control-factor orthogonalisation.

Factor	Ann. ret.	Ann. vol.	Sharpe	HAC t	Max DD	Style R2	Style alpha	Style vol.
gp_to_equity	-2.86%	7.60%	-0.38	-0.63	-22.38%	0.573	-3.19%	7.60%

Factor	MKT	SMB	VAL	MOM	BAB	QMJ	Alpha	R2
gp_to_equity	0.27 (4.43)	-0.00 (-0.08)	-0.32 (-3.35)	-0.17 (-2.11)	0.42 (1.95)	0.43 (5.47)	-2.93% (-1.02)	0.573

Factor	MKT	SMB	VAL	MOM	BAB	QMJ	Alpha	Total
gp_to_equity	5.91%	0.02%	1.62%	-1.71%	-1.08%	-4.69%	-2.93%	-2.86%

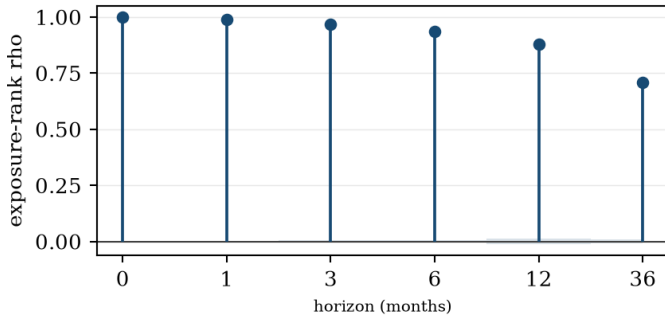


Figure: Exposure-rank ACF; 5% significant only outside the pale HAC band.

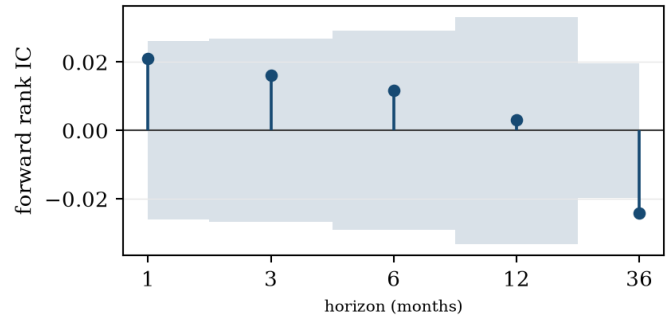


Figure: Exposure-return rank CCF; 5% significant only outside the pale HAC band.

Item	Value
Direction	+1
Common months	41
Signals	gp_to_equity(+1)
Effective construction	rank/rank; 1%-99%; signal_weighted; LS; beta-neutral=True; group=sector; cost=0bp
Components required	1 of 1
Available / comparison months	43 / 41
Available window	2023-01-01 to 2026-07-01
Dropped: low breadth / common	0 / 2
Stocks, min / median / max	1965 / 2331 / 2472
Available-history Sharpe	0.03
Bootstrap Sharpe	-0.38 [-1.48, 0.92]; Pr(S<=0)=73.4%
Bootstrap tails	skew=-0.31 [-0.65, 0.34]; max DD=-22.4% [-37.7%, -3.8%]

17. capex_to_revenue

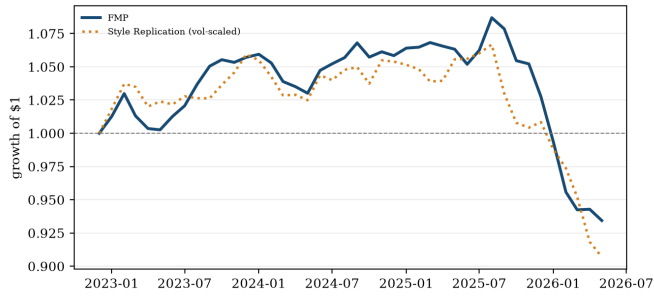


Figure: Common-window cumulative growth of \$1.

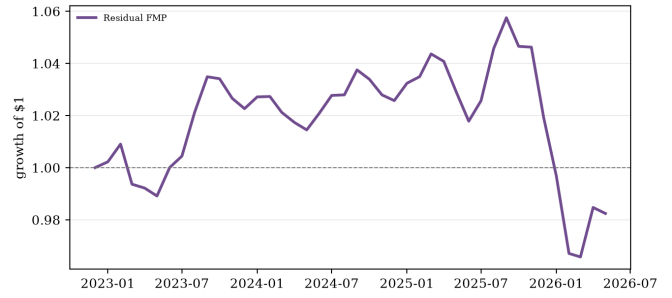


Figure: Cumulative growth after simultaneous control-factor orthogonalisation.

Factor	Ann. ret.	Ann. vol.	Sharpe	HAC t	Max DD	Style R2	Style alpha	Style vol.
capex_to_revenue	-1.88%	4.53%	-0.42	-0.56	-14.04%	0.309	-0.35%	4.53%

Factor	MKT	SMB	VAL	MOM	BAB	QMJ	Alpha	R2
capex_to_revenue	0.06 (0.74)	0.11 (1.56)	-0.22 (-2.85)	-0.17 (-2.55)	0.16 (0.98)	0.11 (0.82)	-0.45% (-0.13)	0.309

Factor	MKT	SMB	VAL	MOM	BAB	QMJ	Alpha	Total
capex_to_revenue	1.28%	-0.55%	1.13%	-1.73%	-0.40%	-1.16%	-0.45%	-1.88%

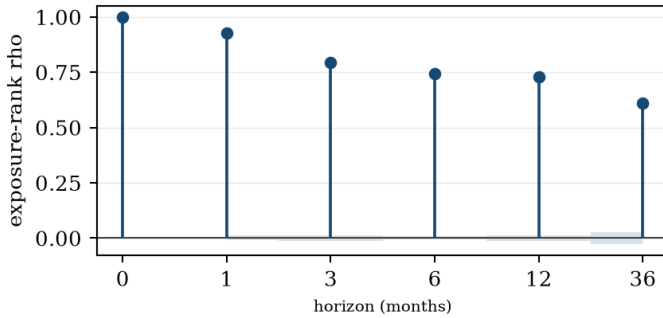


Figure: Exposure-rank ACF; 5% significant only outside the pale HAC band.

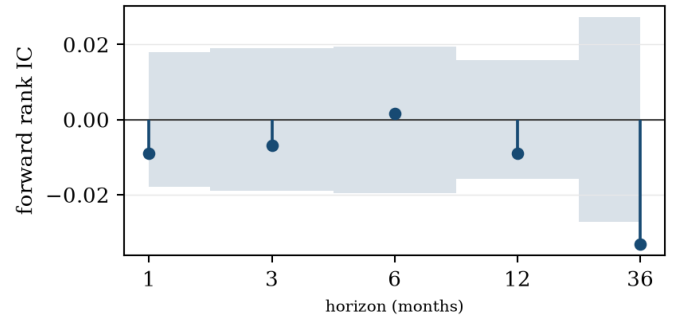


Figure: Exposure-return rank CCF; 5% significant only outside the pale HAC band.

Item	Value
Direction	+1
Common months	41
Signals	capex_to_revenue(-1)
Effective construction	rank/rank; 1%-99%; signal_weighted; LS; beta-neutral=True; group=sector; cost=0bp
Components required	1 of 1
Available / comparison months	43 / 41
Available window	2023-01-01 to 2026-07-01
Dropped: low breadth / common	0 / 2
Stocks, min / median / max	2040 / 2403 / 2619
Available-history Sharpe	-0.18
Bootstrap Sharpe	-0.42 [-1.48, 1.05]; Pr(S<=0)=73.2%
Bootstrap tails	skew=-0.73 [-1.13, 0.23]; max DD=-14.0% [-25.9%, -2.8%]

18. momentum_0_21

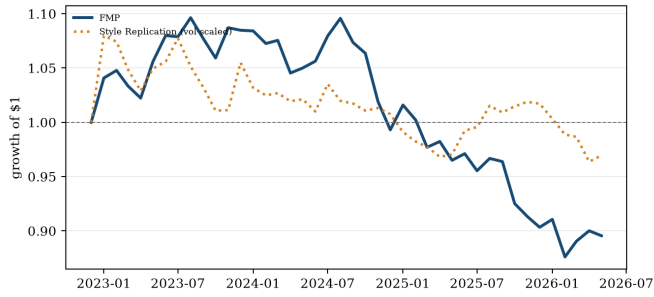


Figure: Common-window cumulative growth of \$1.

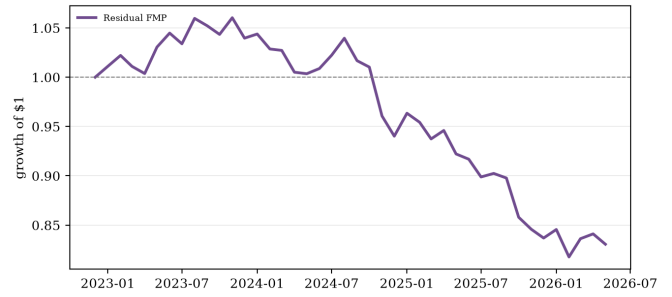


Figure: Cumulative growth after simultaneous control-factor orthogonalisation.

Factor	Ann. ret.	Ann. vol.	Sharpe	HAC t	Max DD	Style R2	Style alpha	Style vol.
momentum_0_21	-3.00%	6.80%	-0.44	-0.87	-20.09%	0.120	-2.76%	6.80%

Factor	MKT	SMB	VAL	MOM	BAB	QMJ	Alpha	R2
momentum_0_21	0.12 (1.28)	0.07 (0.42)	-0.00 (-0.02)	-0.10 (-1.21)	-0.01 (-0.03)	-0.07 (-0.62)	-5.23% (-1.13)	0.160

Factor	MKT	SMB	VAL	MOM	BAB	QMJ	Alpha	Total
momentum_0_21	2.76%	-0.33%	0.01%	-1.01%	0.02%	0.78%	-5.23%	-3.00%

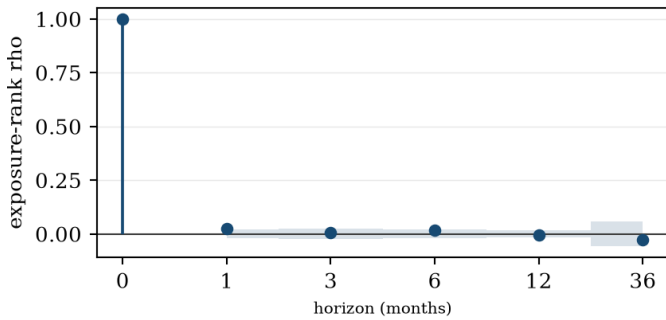


Figure: Exposure-rank ACF; 5% significant only outside the pale HAC band.

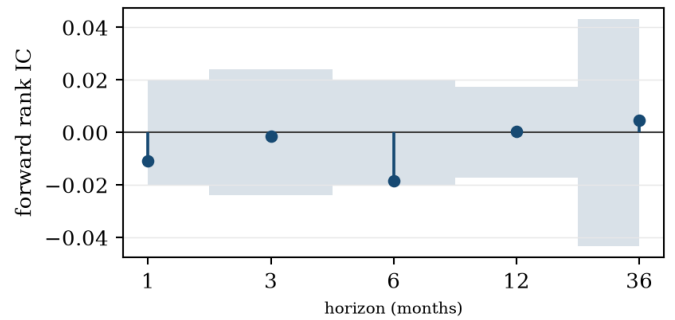


Figure: Exposure-return rank CCF; 5% significant only outside the pale HAC band.

Item	Value
Direction	+1
Common months	41
Signals	momentum_0_21(-1)
Effective construction	rank/rank; 1%-99%; signal_weighted; LS; beta-neutral=True; group=sector; cost=0bp
Components required	1 of 1
Available / comparison months	43 / 41
Available window	2023-01-01 to 2026-07-01
Dropped: low breadth / common	0 / 2
Stocks, min / median / max	2106 / 2547 / 2812
Available-history Sharpe	-0.18
Bootstrap Sharpe	-0.44 [-1.45, 0.59]; Pr(S<=0)=81.3%
Bootstrap tails	skew=0.01 [-0.30, 0.28]; max DD=-20.1% [-29.2%, -5.3%]

19. debt_to_equity

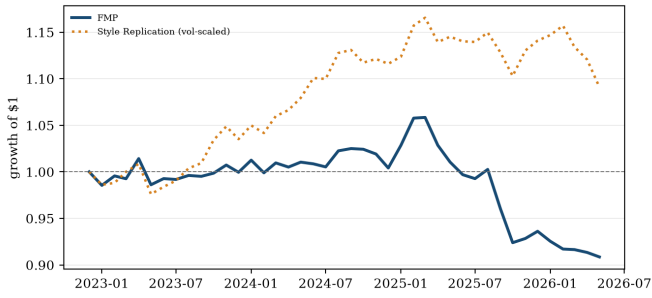


Figure: Common-window cumulative growth of \$1.

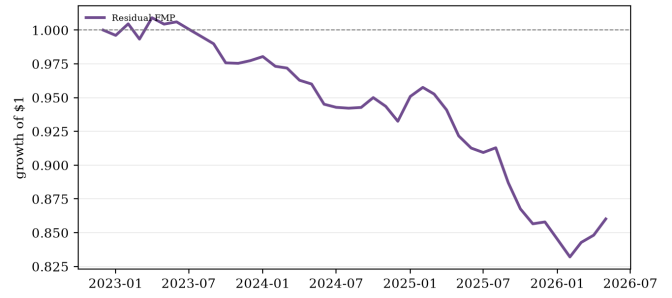


Figure: Cumulative growth after simultaneous control-factor orthogonalisation.

Factor	Ann. ret.	Ann. vol.	Sharpe	HAC t	Max DD	Style R2	Style alpha	Style vol.
debt_to_equity	-2.67%	5.20%	-0.51	-1.04	-14.16%	0.513	-4.59%	5.20%

Factor	MKT	SMB	VAL	MOM	BAB	QMJ	Alpha	R2
debt_to_equity	0.20 (3.39)	-0.09 (-1.08)	-0.03 (-0.34)	-0.08 (-1.45)	0.34 (2.54)	0.16 (1.26)	-4.34% (-2.13)	0.514

Factor	MKT	SMB	VAL	MOM	BAB	QMJ	Alpha	Total
debt_to_equity	4.43%	0.48%	0.15%	-0.81%	-0.86%	-1.72%	-4.34%	-2.67%

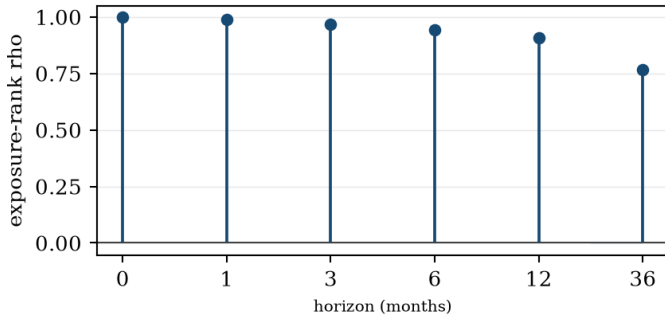


Figure: Exposure-rank ACF; 5% significant only outside the pale HAC band.

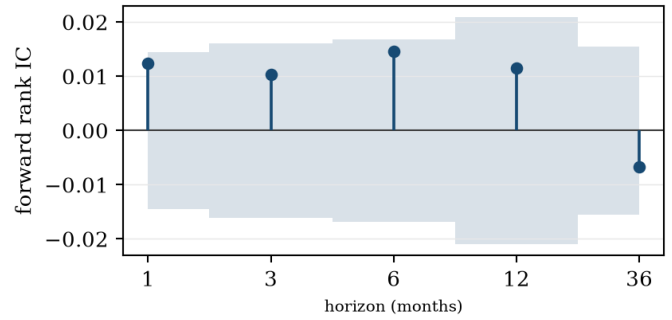


Figure: Exposure-return rank CCF; 5% significant only outside the pale HAC band.

Item	Value
Direction	+1
Common months	41
Signals	debt_to_equity(+1)
Effective construction	rank/rank; 1%-99%; signal_weighted; LS; beta-neutral=True; group=sector; cost=0bp
Components required	1 of 1
Available / comparison months	43 / 41
Available window	2023-01-01 to 2026-07-01
Dropped: low breadth / common	0 / 2
Stocks, min / median / max	1982 / 2384 / 2593
Available-history Sharpe	-0.15
Bootstrap Sharpe	-0.51 [-1.37, 0.53]; Pr(S<=0)=84.9%
Bootstrap tails	skew=-0.56 [-1.18, 0.23]; max DD=-14.2% [-25.9%, -2.8%]

20. price_zscore_63d

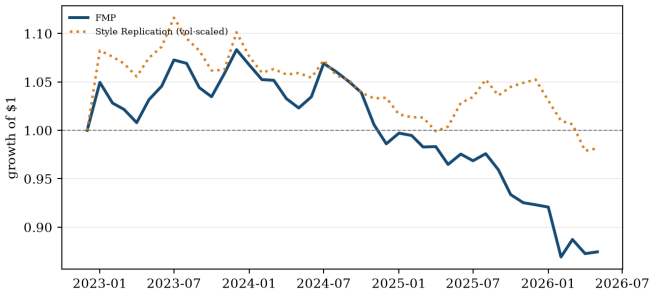


Figure: Common-window cumulative growth of \$1.

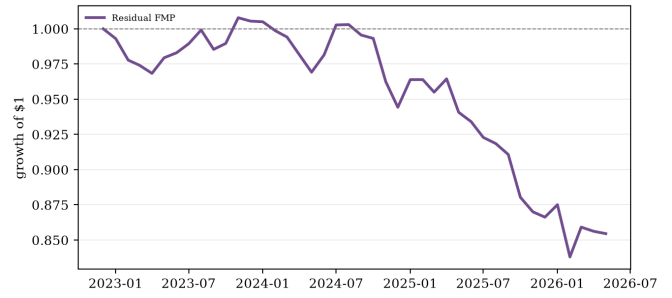


Figure: Cumulative growth after simultaneous control-factor orthogonalisation.

Factor	Ann. ret.	Ann. vol.	Sharpe	HAC t	Max DD	Style R2	Style alpha	Style vol.
price_zscore_63d	-3.69%	6.78%	-0.54	-1.17	-19.76%	0.451	-3.47%	6.78%

Factor	MKT	SMB	VAL	MOM	BAB	QMJ	Alpha	R2
price_zscore_63d	0.15 (1.93)	0.09 (0.83)	0.03 (0.26)	-0.27 (-6.16)	-0.15 (-0.73)	-0.02 (-0.23)	-4.47% (-1.37)	0.458

Factor	MKT	SMB	VAL	MOM	BAB	QMJ	Alpha	Total
price_zscore_63d	3.42%	-0.45%	-0.14%	-2.66%	0.40%	0.22%	-4.47%	-3.69%

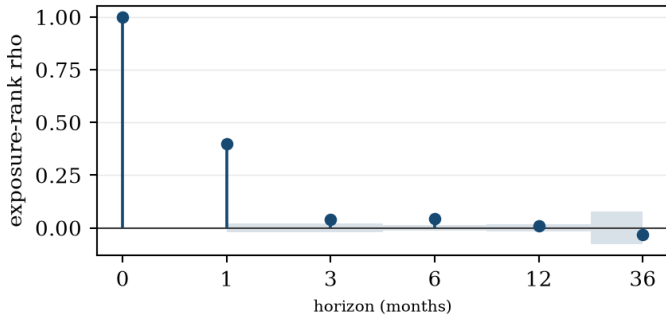


Figure: Exposure-rank ACF; 5% significant only outside the pale HAC band.

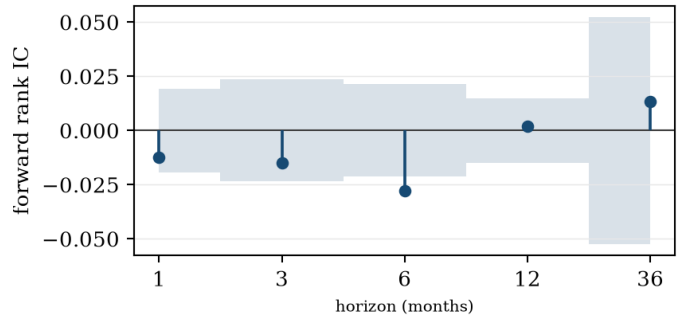


Figure: Exposure-return rank CCF; 5% significant only outside the pale HAC band.

Item	Value
Direction	+1
Common months	41
Signals	price_zscore_63d(-1)
Effective construction	rank/rank; 1%-99%; signal_weighted; LS; beta-neutral=True; group=sector; cost=0bp
Components required	1 of 1
Available / comparison months	43 / 41
Available window	2023-01-01 to 2026-07-01
Dropped: low breadth / common	0 / 2
Stocks, min / median / max	2106 / 2547 / 2811
Available-history Sharpe	-0.26
Bootstrap Sharpe	-0.54 [-1.46, 0.28]; Pr(S<=0)=90.7%
Bootstrap tails	skew=0.28 [-0.39, 0.95]; max DD=-19.8% [-28.1%, -6.9%]

21. ma_dist_63d

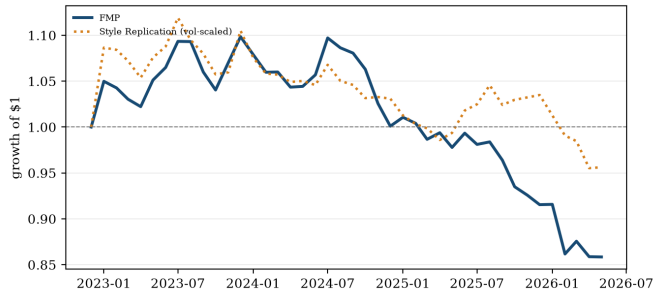


Figure: Common-window cumulative growth of \$1.

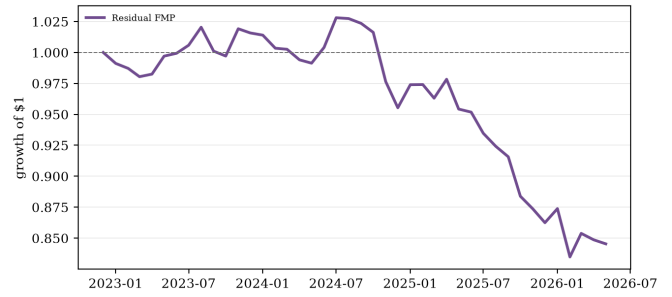


Figure: Cumulative growth after simultaneous control-factor orthogonalisation.

Factor	Ann. ret.	Ann. vol.	Sharpe	HAC t	Max DD	Style R2	Style alpha	Style vol.
ma_dist_63d	-4.19%	7.38%	-0.57	-1.10	-21.85%	0.441	-3.49%	7.38%

Factor	MKT	SMB	VAL	MOM	BAB	QMJ	Alpha	R2
ma_dist_63d	0.16 (1.72)	0.16 (1.24)	-0.01 (-0.11)	-0.28 (-6.51)	-0.13 (-0.61)	-0.01 (-0.06)	-4.76% (-1.22)	0.450

Factor	MKT	SMB	VAL	MOM	BAB	QMJ	Alpha	Total
ma_dist_63d	3.66%	-0.79%	0.06%	-2.76%	0.33%	0.07%	-4.76%	-4.19%

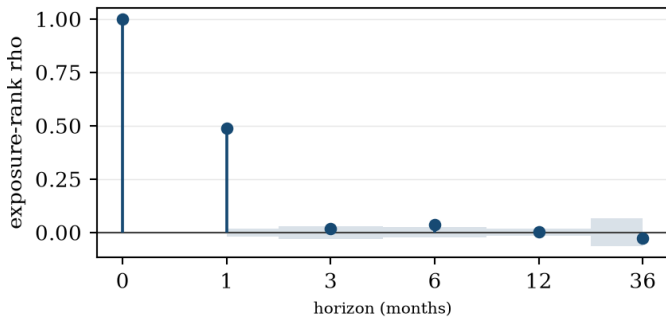


Figure: Exposure-rank ACF; 5% significant only outside the pale HAC band.

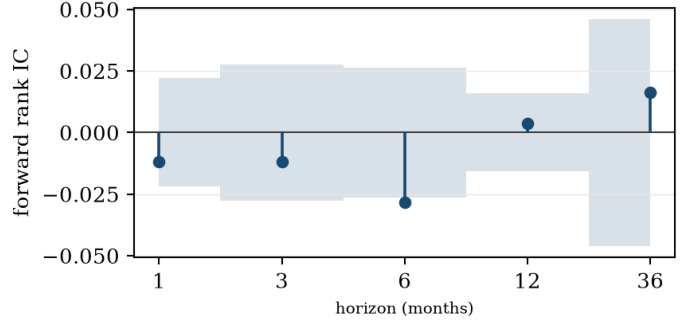


Figure: Exposure-return rank CCF; 5% significant only outside the pale HAC band.

Item	Value
Direction	+1
Common months	41
Signals	ma_dist_63d(-1)
Effective construction	rank/rank; 1%-99%; signal_weighted; LS; beta-neutral=True; group=sector; cost=0bp
Components required	1 of 1
Available / comparison months	43 / 41
Available window	2023-01-01 to 2026-07-01
Dropped: low breadth / common	0 / 2
Stocks, min / median / max	2106 / 2547 / 2812
Available-history Sharpe	-0.28
Bootstrap Sharpe	-0.57 [-1.53, 0.33]; Pr(S<=0)=89.5%
Bootstrap tails	skew=0.25 [-0.27, 0.81]; max DD=-21.8% [-31.9%, -7.0%]

Reproducibility

Run ID: 20260906T115831Z_no-clustering-bootstrap-screen-sharpe-0-5-oos-2023_b9d1626e52.

Gold snapshot: version=20260829T051952Z.

Configuration hash: b9d1626e52.

Symbols requested / loaded: 0 / 3595.

Seed: 0. Comparison months: 41, 2023-01-01 to 2026-05-01.

Command:

```
C:/Users/User/Desktop/alpha/.venv/Scripts/python.exeC:/Users/User/Desktop/alpha/src/experiments/automatic_factor_grouping/screen_handoff.py--screen-sourcedata/outputs/automatic_factor_grouping/signal_screening/20260906T064949Z_e2fdcf7479--panel-sourcedata/outputs/automatic_factor_grouping/runs/20260906T065013Z_25145edccf--cutoff0.5.
```

Direction is frozen upstream; this report performs no return-based sign selection. The reported direction is applied consistently before every downstream regression, style calculation, plot, table, PCA coordinate, and index assignment. The run directory contains the source configuration, normalized configuration, manifest, canonical tables, monthly returns, PCA coordinates, plots, report source, and PDF.